



ACQLERATE

MODULE 2

Defense Finance & Budgeting

Master the PPBE cycle, appropriations law, EVM, and financial management fundamentals that every PM must know.

10 Lessons · ~245 Minutes · Lesson Book Edition

Course Lessons

01	The PPBE Cycle	15 min
02	Color of Money by Service	20 min
03	Cost Estimating	24 min
04	Earned Value Management (EVM)	18 min
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DEFENSE FINANCE & BUDGETING · LESSON 1

The PPBE Cycle

15 min

Understand how DoD programs are funded through the Planning, Programming, Budgeting, and Execution process.

What PPBE Actually Is

PPBE stands for Planning, Programming, Budgeting, and Execution. It is the DoD's annual process for deciding what gets funded, and it runs on roughly a two year cycle from first identifying a need to Congress actually appropriating the money. Every dollar your program gets has to survive all four phases. Your program's funding is not automatic. Every year, program managers must compete for resources within the PPBE process. Understanding this cycle: who makes decisions, when, and based on what criteria: is the difference between a well-funded program and one that gets cut or restructured. Missing a POM submission window can delay your program by 2 years.

How Money Flows from Strategy to Your Program

The PPBE Annual Cycle

THE BENNY HILL PROBLEM

Programs that don't have a well-documented funding justification going into the POM get "Benny Hill'd": cut to fund higher priorities. Your job as PM is to have a compelling cost/benefit narrative, performance data, and risk arguments ready every time OSD reviews your program. Document everything.

Appropriations Types You'll Manage

- **RDT&E (Research, Development, Test & Evaluation), 3-year availability; used to develop and test new systems** RDT&E has six budget activities: 6.1 (Basic Research), 6.2 (Applied Research), 6.3 (Advanced Technology Development), 6.4 (Advanced Component Development), 6.5 (System Development & Demonstration), 6.6 (Management Support). The budget activity number appears in all program budget exhibits and determines what specific development work is authorized.
- **Procurement, 3-year availability; used to buy production-ready systems** Procurement funds buy end items in production quantities: aircraft, vehicles, ships, weapons. They cannot be used for development work (that's RDT&E) or for sustaining fielded systems (that's O&M). The 3-year availability means FY25 Procurement funds can obligate new contracts through September 2027, then enter a 5-year expired period for payment of existing obligations only.

- **O&M (Operations & Maintenance), 1-year availability; day-to-day operating costs and services** O&M is the workhorse appropriation: it pays for everything from base operations to most services contracts to contractor maintenance. The 1-year obligation period creates intense fiscal year-end pressure. The \$350K expense/investment threshold also matters (raised from \$250K: DAU's own Acquipedia now cites \$350K): O&M cannot be used for construction that exceeds the MILCON threshold (generally \$1.5M+).
- **MILCON (Military Construction), 5-year availability; construction of facilities** MILCON funds permanent construction, renovation, and real property acquisition. The 5-year availability reflects long project timelines. Using O&M for construction above MILCON thresholds is a common ADA violation: the line between "maintenance/repair" (O&M) and "construction" (MILCON) is frequently contested and often wrong in practice.
- **MILPERS (Military Personnel), 1-year; pay and allowances for uniformed personnel** MILPERS pays active duty military salaries, housing allowances (BAH), subsistence (BAS), hazardous duty pay, and separation pay. It cannot fund civilian positions (that's O&M) or contractor services. MILPERS accounts are among the most tightly controlled in DoD because Congress is extremely sensitive to military compensation issues.

Monthly Burn Rate Formula

ANTI-DEFICIENCY ACT

Never obligate funds in excess of what is appropriated or in advance of appropriation. Violations of the Anti-Deficiency Act (31 U.S.C. §§ 1341, 1342) are federal crimes and career-ending events. Every PM must understand these constraints cold.

INTERMEDIATE

POM Execution: Managing Your Program Through the Budget Cycle

Mid-career PMs live in two worlds simultaneously: executing the current year's approved budget and defending next year's POM submission. Your POM submission must be defensible with cost data, supported by your program's APB, and internally consistent with your contract structure. The most common POM mistake: projecting out-year costs that don't match your contract's performance schedule, or requesting procurement funding before RDT&E work is complete. Budget analysts at the Component and OSD level will catch these inconsistencies. Build your POM from the contract up. Not from the budget down. Cross-walk your WBS to your budget line items. When the two don't align, you get reprogramming problems in execution.

ADVANCED**CONGRESSIONAL MARKS AND UNFUNDED PRIORITIES: THE HIDDEN BUDGET GAME**

The President's Budget Request is a starting position, not an ending position. Congress marks up every defense bill. Adding earmarks, cutting programs, fencing funds pending reports. Senior PMs must track their program's budget through the NDAA markup process: House HASC mark, Senate SASC mark, conference report. Fenced funding (congressionally restricted) requires notifications or reports before obligation. Failing to track these leads to Anti-Deficiency Act violations. Unfunded Priorities Lists (UPLs) are the Service Chiefs' mechanism for asking Congress to add money above the budget request. If your program is on a UPL, it creates both opportunity (more money) and risk (congressional attention). Know where your program sits in the congressional priority stack. Your program's survival may depend on it.

Key Terms**PPBE**

Planning, Programming, Budgeting, and Execution. DoD's annual resource management process.

POM

Program Objectives Memorandum. The Services' funding proposals submitted to OSD.

FYDP

Future Years Defense Program, 5-year funding baseline for all DoD programs.

PBD

Program Budget Decision. OSD decisions on Service POMs that shape the President's Budget.

NDAA

National Defense Authorization Act. Annual legislation authorizing DoD activities and funding.

Knowledge Check

What is the Future Years Defense Program (FYDP)?

- A. A 10-year strategic plan
- ✓ **B. A 5-year resource database tracking all DoD programs and funding**
- C. The Congressional budget resolution for defense
- D. The Services' annual programming document submitted to OSD

WHY

The FYDP is the official DoD database that tracks all programs and resources over a 5-year period. It serves as the baseline for the annual PPBE process and is the authoritative source for program funding profiles.

O&M (Operations & Maintenance) appropriations have how many years of availability?

- ✓ **A. 1 year**
- B. 2 years
- C. 3 years
- D. 5 years

WHY

O&M funds have a 1-year period of availability. This means they must be obligated within the fiscal year they are appropriated. Unused O&M funds expire at year-end and cannot be used to fund future activities.

Violating the Anti-Deficiency Act by obligating funds in excess of appropriations can result in:

- A. A formal letter of reprimand
- ✓ **B. Administrative, civil, and criminal penalties up to imprisonment**
- C. Program restructuring only
- D. A 30-day funding freeze

WHY

The Anti-Deficiency Act (31 U.S.C. §§ 1341, 1342) violations can result in administrative discipline, civil fines, and criminal prosecution. In practice, they are career-ending events that are taken extremely seriously at all levels of DoD.

Which document do the Military Services submit to OSD during the Programming phase to propose how they plan to allocate resources across programs?

A. Budget Estimate Submission (BES)

✓ **B. Program Objectives Memorandum (POM)**

C. Future Years Defense Program (FYDP)

D. Defense Planning Guidance (DPG)

WHY

The Program Objectives Memorandum (POM) is the Services' formal proposal to OSD during the Programming phase of PPBE. It outlines how the Service proposes to allocate resources across programs over the FYDP. OSD then issues Program Budget Decisions (PBDs) to adjust the POM.

What is the period of availability for Procurement appropriations?

A. 1 year

B. 2 years

✓ **C. 3 years**

D. 5 years

WHY

Procurement appropriations have a 3-year period of availability. This allows programs to obligate production contracts over multiple fiscal years. After the 3-year availability period, unobligated funds enter a 5-year expired period before being cancelled.

What happens to appropriated funds that are not obligated before the end of their period of availability?

A. They are immediately cancelled and returned to Treasury

✓ **B. They enter an "expired" status for 5 years, then are cancelled**

C. They roll over automatically to the next fiscal year

D. They convert to O&M funds for current operations

WHY

Unobligated funds that pass their period of availability enter a 5-year expired period. During this time, they can still be used to adjust or liquidate existing obligations but cannot fund new obligations. After 5 years, the funds are cancelled and returned to the Treasury.

The Defense Planning Guidance (DPG) is issued during which phase of PPBE and by whom?

- A. Budgeting phase, by OMB
- B. Execution phase, by the Comptroller

✓ C. Planning phase, by the Secretary of Defense

- D. Programming phase, by the Service Secretaries

WHY

The DPG is issued by the Secretary of Defense during the Planning phase (approximately 2 years before execution). It sets strategic priorities and guidance that shape how Services structure their POMs. Programs not aligned to DPG priorities risk being unfunded.

A program office under-executing its O&M budget late in the fiscal year ("low burn rate") risks what consequence?

- A. Immediate funding cancellation
- B. A supplemental appropriation request

✓ C. Reduced future-year funding due to apparent lack of need

- D. An automatic contract extension

WHY

Under-execution signals to higher headquarters that the program doesn't need as much money as requested, leading to reductions in future-year FYDP funding. The "use it or lose it" dynamic in O&M encourages prudent year-end execution but can lead to poor spending decisions if not managed carefully.

MILCON (Military Construction) appropriations have what period of availability?

- A. 1 year
- B. 2 years
- C. 3 years

✓ D. 5 years

WHY

Military Construction (MILCON) appropriations have a 5-year period of availability, reflecting the lengthy nature of construction projects. This is the longest standard period among the major appropriation types.

Which congressional action authorizes DoD activities and programs but does NOT actually provide spending authority?

A. Appropriations Act

✓ **B. National Defense Authorization Act (NDAA)**

C. Continuing Resolution

D. Omnibus Spending Bill

WHY

The NDAA authorizes programs, policy, and force structure but does NOT appropriate money. A separate annual Appropriations Act (or Omnibus bill) provides the actual spending authority. Programs can be authorized but not funded if Congress passes an NDAA without a corresponding appropriation.

Place the PPBE phases in the correct order as they occur in the annual DoD budget cycle:

CORRECT ORDER

1. Planning: Issue DPG strategic guidance
2. Programming: Services submit Program Objectives Memorandum (POM)
3. Budgeting: OSD/OMB review; President's Budget submitted to Congress
4. Execution: Obligate and expend appropriated funds

WHY

The PPBE cycle flows: Planning (strategic guidance) → Programming (POM/RMD development) → Budgeting (President's Budget submission) → Execution (obligation and expenditure of appropriated funds). Each phase feeds the next; execution data feeds back into next year's planning.

Match each appropriation type to its standard period of availability:

CORRECT MATCHES

O&M (Operations & Maintenance)	1 Year
RDT&E (Research, Dev, Test & Eval)	2 Years
Procurement	3 Years
MILCON (Military Construction)	5 Years

WHY

Knowing the period of availability is critical for PMs: O&M is 1-year (use it or lose it each fiscal year), Procurement is 3-year (for major equipment), RDT&E is 2-year (research and development), and MILCON is 5-year (construction projects take time).

DEFENSE FINANCE & BUDGETING · LESSON 2

Color of Money by Service

20 min

Master the specific appropriation accounts used across the Army, Navy, Air Force, and Marine Corps. And why using the wrong "color of money" is a federal violation.

Why "Color of Money" Can End Your Career

Every appropriation type has strict statutory limits on what it can buy. Using RDT&E money to fund a production contract, or O&M money to fund a capital acquisition, is a federal violation of the Purpose Statute (31 U.S.C. § 1301). Program Managers who ignore these rules face potential Antideficiency Act violations, personal liability, and career termination. Understanding each appropriation's "color": and matching the right color to the right expenditure: is non-negotiable.

Major Appropriation Types: Rules & Period of Availability

The Three Laws Governing Appropriations Use

Three statutory principles govern every spending decision in DoD: (1) The Purpose Statute (31 U.S.C. § 1301): funds may only be used for what Congress intended. (2) The Time Statute (31 U.S.C. § 1502): funds may only obligate for needs arising within their period of availability. (3) The Amount Statute / Antideficiency Act (31 U.S.C. §§ 1341, 1342): cannot obligate more than was appropriated. Violating any of these three is a federal offense.

REAL TALK

A table of accounting codes is exactly as thrilling as it sounds, and I won't pretend otherwise. Here's why you should still glance at it: your finance team books every dollar on your program against one of these codes, and if a customer ever asks you a funding question, knowing these codes exist and differ by service keeps you from an awkward silence. One thing actually worth remembering: an Army code will not carry over to a Navy program. Otherwise, skim it and move on with your life.

Per-Service Treasury Account Symbols (TAS): Key Accounts

THE AF 3400 ACCOUNT: A PM'S DAILY REALITY

Air Force 3400 (O&M) is the most frequently used appropriation in day-to-day AF program management. It funds training, logistics, maintenance, most advisory & assistance services, and sustainment contracts. When PMs confuse 3400 (O&M) with 3600 (RDT&E): for example, using 3400 to pay for developmental testing: they violate the Purpose Statute. Always ask: "Is this work developing/testing a new capability (RDT&E), buying a production item (Procurement), or sustaining/operating an existing capability (O&M)?"

Color of Money: Which Appropriation?

What Happens When Funds Expire and Cancel

After a fund's period of availability ends, unobligated balances enter "expired" status for 5 years. During this period, the funds can still be used to adjust existing obligations (e.g., pay a contract invoice that was obligated on time). After the 5-year expired window, funds are permanently "cancelled" and returned to the Treasury: they cannot be used for any purpose. A FY2024 O&M obligation that generates an invoice in FY2028 is still payable; a FY2024 O&M invoice arriving in FY2031 cannot be paid from those funds.

REAL EXAMPLE: BUYING A RADAR SYSTEM

A new radar program follows this funding progression: Basic research uses RDT&E (3600 for AF). Engineering development and testing uses RDT&E. First production units use Procurement (3080 for AF). Fielded radar maintenance and upgrades use O&M (3400 for AF). Training operators uses O&M. Building a radar maintenance facility uses MILCON. The same physical system touches four different appropriation accounts over its life.

INTERMEDIATE

Managing Color of Money Across a Complex Program

Real programs use multiple appropriation types simultaneously. A typical ACAT II program in EMD might have RDT&E funding the development contract, O&M funding government lab support, and MILCON funding a test facility. Each appropriation has different obligation and expenditure rules. The most common mid-career mistake: using RDT&E to fund activities that have crossed into production (violates the appropriation's purpose), or letting O&M funds expire while trying to transfer them to RDT&E (not permitted). Your comptroller is your best friend. But you need to give them enough lead time to structure funding actions properly. Last-minute funding transfers almost always fail or create audit exposure.

ADVANCED**ANTI-DEFICIENCY ACT VIOLATIONS AND HOW THEY HAPPEN TO GOOD PMS**

The Anti-Deficiency Act (31 U.S.C. § 1341) prohibits obligating funds in excess of or in advance of appropriations. Violations are career-ending. They happen most often through: (1) continuing contract performance after a CR expires and before new appropriations are enacted without a proper Continuing Resolution Authority authorization; (2) using one appropriation to fund activities that belong to another ("purpose" violations); (3) obligating contracts before funds are available in USASpending. Senior PMs must build a personal funding calendar: when does each appropriation expire? When does the CR end? What is the period of availability for each line item? These are not comptroller problems. They are PM problems. The PM signs the obligation documents.

Key Terms**TAS**

Treasury Account Symbol. The unique identifier for each appropriation account (e.g., AF 3400 for Air Force O&M).

O&M

Operations & Maintenance, 1-year appropriation funding day-to-day operations, maintenance, and most services contracts.

RDT&E

Research, Development, Test & Evaluation, 2-year appropriation funding research and development activities.

Procurement

3-year appropriation funding production and purchase of end-items (aircraft, ships, missiles, vehicles).

MILCON

Military Construction, 5-year appropriation funding construction of military facilities.

Bona Fide Need Rule

Funds may only be used to satisfy a legitimate, genuine need that arose during the period of availability.

Purpose Statute

31 U.S.C. § 1301. Appropriations may only be applied to the objects for which they were made.

Color of Money

The appropriation type funding a particular contract line. Mixing appropriation types on one CLIN is illegal.

Expired Funds

Funds past their period of availability but still available for 5 years to adjust or pay existing obligations.

Canceled Funds

Funds that have completed the 5-year expired period; permanently returned to the Treasury.

Knowledge Check

What is the Air Force Treasury Account Symbol (TAS) for Operations & Maintenance funds?

- A. 57-3600
- B. 57-3010
- ✓ C. 57-3400
- D. 57-3080

WHY

The Air Force O&M appropriation is identified by TAS 57-3400 (commonly called "3400 money"). It has a 1-year period of availability and funds day-to-day AF operations, training, maintenance, and most services contracts. 3600 is RDT&E, 3010 is Aircraft Procurement, and 3080 is Other Procurement.

Which statutory principle states that appropriations may only be used for the purposes Congress intended?

- A. Anti-Deficiency Act
- B. Bona Fide Need Rule
- ✓ C. Purpose Statute (31 U.S.C. § 1301)
- D. Time Statute (31 U.S.C. § 1502)

WHY

The Purpose Statute (31 U.S.C. § 1301) is the core rule that each appropriation may only be applied to its intended purpose. Using O&M funds for procurement, or RDT&E for O&M activities, violates this statute: regardless of whether sufficient funds exist.

RDT&E appropriations have what period of availability?

A. 1 year

✓ B. 2 years

C. 3 years

D. 5 years

WHY

RDT&E appropriations have a 2-year period of availability, reflecting the multi-year nature of development activities. O&M has 1 year, Procurement has 3 years, and MILCON has 5 years.

The Navy's Shipbuilding & Conversion (SCN) appropriation uses which Treasury Account Symbol?

A. 17-1506

B. 17-1507

✓ C. 17-1611

D. 17-1804

WHY

Navy Shipbuilding & Conversion (SCN) uses TAS 17-1611. This is the appropriation that funds construction of Navy vessels: carriers (CVN), destroyers (DDG), submarines (SSN/SSBN), and amphibious ships. 17-1506 is Aircraft Procurement Navy, 17-1507 is Weapons Procurement Navy, and 17-1804 is Other Procurement Navy.

An Army program wants to buy 500 Javelin anti-tank missile systems in production. Which appropriation should be used?

A. Army O&M (2400)

B. Army RDT&E

✓ C. Army Missile Procurement (21-2020)

D. Army Other Procurement (21-2060)

WHY

Production quantities of guided missiles are funded with Procurement appropriations. For the Army, Missile Procurement (TAS 21-2020) specifically funds production of missiles including Javelin, Patriot, and Stinger. Using O&M or RDT&E for production purchases would violate the Purpose Statute.

The Bona Fide Need Rule requires that appropriated funds be used for:

A. The most cost-effective solution available

✓ **B. Needs that legitimately arose during the appropriation's period of availability**

C. Purchases over \$350K only

D. Programs listed in the FYDP

WHY

The Bona Fide Need Rule (31 U.S.C. § 1502(a)) requires that funds obligated must meet a legitimate need that arose during the fund's period of availability. For example, you cannot use FY2025 O&M funds in FY2025 to pre-pay for services entirely to be delivered in FY2027: the need hasn't arisen yet.

After a fund's period of availability expires, the unobligated balance enters "expired" status. For how many additional years can these expired funds still be used to pay existing obligations?

A. 1 year

B. 2 years

✓ **C. 5 years**

D. They cannot be used at all

WHY

After the period of availability ends, funds enter a 5-year expired period during which they can still adjust or liquidate existing obligations (pay invoices on contracts that were properly obligated during the availability period). After 5 years, the funds are permanently cancelled and returned to the Treasury.

Which Air Force appropriation specifically funds the procurement of aircraft such as the F-35A and KC-46?

A. 57-3400 (O&M)

B. 57-3600 (RDT&E)

✓ **C. 57-3010 (Aircraft Procurement)**

D. 57-3080 (Other Procurement AF)

WHY

TAS 57-3010: Aircraft Procurement, Air Force: funds the production purchase of Air Force aircraft including the F-35A, KC-46 tanker, C-130J, and B-21. RDT&E (3600) funds their development and testing, while O&M (3400) funds sustainment after fielding.

A program manager uses O&M funds to construct a new \$2 million maintenance facility. This likely violates:

- A. The Bona Fide Need Rule only
- B. The Time Statute only

✓ C. The Purpose Statute: construction over \$750K threshold requires MILCON appropriations

- D. No rule, since O&M can fund any maintenance-related activity

WHY

Permanent construction above \$1.5M (\$750K in some contexts) generally requires MILCON appropriations: not O&M. Using O&M for construction that should be MILCON violates the Purpose Statute. "Minor construction" below threshold may use O&M, but a \$2M facility exceeds that threshold.

DARPA's research and development funding falls under which Defense-Wide appropriation?

- A. 97-0400 (O&M Defense-Wide)

✓ B. 97-0400 (RDT&E Defense-Wide)

- C. 17-1506 (Aircraft Procurement Navy)
- D. 57-3600 (RDT&E Air Force)

WHY

DARPA, the Missile Defense Agency (MDA), and other defense-wide R&D activities are funded through RDT&E Defense-Wide appropriations (TAS 97-0603/0400 series). Each Service has its own RDT&E account (Air Force 3600, Army 2040, Navy 1319), but DARPA and other OSD-level agencies use the Defense-Wide account.

DEFENSE FINANCE & BUDGETING · LESSON 3

Cost Estimating

24 min

Master cost estimating methodologies, the CAPE independent cost estimate process, and. Critically. How contractors price labor through LCATs, pre-priced labor, job codes, and compensation ratios that directly drive your program cost.

Why Cost Estimates Matter in DoD

Cost growth is the number one reason programs get cancelled or restructured. The DoD has struggled historically with "optimism bias": the tendency to underprice programs to win approval, then face costly breaches later. Understanding estimating methodologies helps PMs challenge unrealistic estimates and build credible baselines from day one.

The Three Primary Estimating Methods

- **Analogous (Top-Down): Uses historical data from similar programs. Fast but less accurate. Best for early phase estimates ($\pm 50\%$).** Analogous estimating takes a completed similar program and adjusts for size, complexity, and technology differences. RAND's DAPCA (Development and Production Cost Analysis) models are classic examples. Accuracy is typically $\pm 30\text{-}50\%$: sufficient for budget formulation but not for contract negotiations. CAPE relies heavily on analogous methods for ICEs.
- **Parametric: Uses Cost Estimating Relationships (CERs): statistical models built from large databases. Moderately accurate. Best for early-mid phase ($\pm 25\%$).** CERs express cost as a mathematical function of physical or performance parameters: e.g., $\text{cost} = f(\text{weight, speed, TRL})$. The SEER and PRICE parametric models are widely used in DoD. Parametric estimates can achieve $\pm 15\text{-}25\%$ accuracy when calibrated databases exist. They are particularly useful for novel systems where no single good analog exists.
- **Engineering Build-Up (Bottoms-Up): Detailed task-by-task estimate. Most accurate but requires mature design. Best for milestone B+ ($\pm 10\text{-}15\%$).** Bottoms-up estimates require a detailed WBS, labor category identification, hours per task, rates, and material costs. Accuracy can approach $\pm 5\text{-}10\%$ for mature programs. Contractors use bottoms-up for their proposals; government uses it to build the Independent Government Cost Estimate (IGCE). Requires significant estimating resources: typically 2-6 months for a major system.
- **Hybrid approaches: Most programs use a combination: parametric for unknowns, bottoms-up for well-defined work.** A typical major acquisition uses bottoms-up for well-understood subsystems (existing designs), parametric for new development (using CERs calibrated to similar past programs), and analogous for system-level sanity checks. The estimating methodology must be documented and defensible to DCAA and CAPE review.

THE ROLE OF CAPE

The Cost Assessment and Program Evaluation (CAPE) office, within OSD, provides independent cost assessments for major defense acquisition programs. Their estimates are typically higher than program office estimates: and statistically, CAPE is more often correct. When CAPE's estimate exceeds the program office estimate by 20%+, Congress and leadership take note.

Cost Risk Calculation**NUNN-MCCURDY**

The Nunn-McCurdy Act requires DoD to notify Congress when program unit costs breach specific thresholds (15% = "significant breach"; 25% = "critical breach"). A critical breach requires the program to be certified by USD(A&S) or face cancellation. As a PM, preventing a Nunn-McCurdy breach is a top priority.

Labor Categories (LCATs): The Hidden Engine of Cost Proposals

When a contractor submits a cost proposal, the single largest line item: typically 50-70% of total contract cost: is direct labor. That labor is not proposed as a lump sum. It is built from Labor Categories (LCATs): defined, titled skill levels that specify the education, years of experience, and functional role a person must have to work in that slot. Examples: "Systems Engineer Level II (SE-II)" requires a BS in engineering + 5-8 years experience. "Program Manager Level III (PM-III)" requires 10+ years managing programs > \$50M. Each LCAT has a specific direct labor rate: the hourly base wage: that is burdened with fringe, overhead, and G&A before you see the final billing rate. As a PM, your ability to evaluate whether a contractor's proposed labor mix is realistic directly determines whether your cost estimate is credible.

LCAT Structure: How a Contractor Builds a Labor Proposal**PRE-PRICED LCATS: HOW IDIQS LOCK IN LABOR RATES**

On IDIQ contracts and GWACs (like SEAPORT-NXG, OASIS, GSAM), labor rates are negotiated and fixed in the base contract: these are called pre-priced labor categories. When a task order is competed under that vehicle, the contractor cannot re-price their LCATs above the contract ceiling rates. This is powerful for the government: you are comparing proposals on the same rate structure, so competition becomes about hours and approach rather than rate games. As a PM, when you use a pre-priced IDIQ, verify the ceiling rates were set competitively and recently: rates negotiated 5 years ago may no longer reflect the labor market, and contractors may propose fewer hours at higher mix levels to compensate.

Compa-Ratio: Labor Rate Health Check

How LCATs Connect to Your Cost Estimate as a PM

When you review a contractor's cost proposal at Milestone B or for a new task order, don't just look at total price. Drill into the labor mix. Ask: What percentage of hours are at senior LCAT levels vs. junior? Is the proposed mix realistic for the work? A software sustainment task that is 80% "Software Engineer III" when "Software Engineer I and II" could do 60% of the work is a red flag: it inflates cost without improving quality. Also watch for "labor substitution" clauses in contracts: if the contract requires senior engineers but the contractor is actually staffing junior personnel (billing at junior rates but billing more hours), your actual cost stays the same but quality drops. The LCAT structure, job codes, and compensation ratios are DCAA's primary audit tools to catch exactly this kind of misbilling; and they should be YOUR primary evaluation tools when reviewing proposals.

Five Questions Every PM Should Ask When Reviewing a Labor Proposal

- **1. Does the LCAT mix make sense for the statement of work? Senior engineers should not dominate routine maintenance tasks.** Labor Category (LCAT) mixing is one of the most common areas of over-proposal. A contractor proposing 80% senior engineers for help desk work is either building in margin or doesn't understand the requirement. The government should map each LCAT to specific SOW tasks and challenge any LCAT that doesn't align to the work described.
- **2. Are the proposed direct labor rates consistent with the contractor's disclosed forward pricing rates (FPRAs)? If not, why not?** DCAA negotiates Forward Pricing Rate Agreements (FPRAs) with major contractors annually. These rates should be the ceiling for proposed direct labor. If a contractor proposes rates above their FPRA without explanation, that is a negotiation issue. Small businesses without FPRAs should be compared to market data (e.g., Bureau of Labor Statistics, salary surveys).
- **3. For IDIQ/pre-priced vehicles: are the proposed rates at or below the contract ceiling rates? Any ceiling rate exception requires CO approval.** On pre-priced IDIQ task orders (where rates were established at vehicle award), the ceiling rates in the base contract are the maximum allowable. Proposing above ceiling rates is non-compliant and technically unacceptable: the CO should reject the proposal or require rate conformance before award.
- **4. Is the contractor proposing the same LCATs they actually staff? Request key personnel resumes and verify job codes match proposed LCATs.** Bait-and-switch is one of the most persistent problems in defense services contracting: companies propose impressive senior personnel to win, then staff with less qualified individuals. Key personnel clauses (requiring government approval for substitution) are a partial mitigation. Checking that proposed LCATs match the actual staffing plan is essential due diligence.
- **5. What is the annual escalation rate applied to labor in out-years? Defense labor typically escalates 3-4% annually: proposals using 0% escalation are understating future costs.** Labor escalation compounds significantly over multi-year contracts. A 5-year contract with 4% annual escalation on \$10M year 1 labor reaches \$12.2M in year 5: a 22% increase. The government should verify escalation assumptions

against published indices (Employment Cost Index, Defense-specific labor surveys) and challenge escalation rates above historical norms.

INTERMEDIATE

Defending Your Cost Estimate in a CAPE Review

The OSD Cost Assessment and Program Evaluation (CAPE) office produces Independent Cost Estimates (ICEs) for major programs at each milestone. When CAPE's ICE is significantly higher than your Program Office Estimate (POE), the MDA must choose which to use for the APB. And MDA almost always uses the higher one. Mid-career PMs need to understand how CAPE builds its ICEs: parametric models (SEER, PRICE, COCOMO for software) calibrated to historical analogues. To reduce the gap, you must document your differences with CAPE: what makes your program different from the analogues, what risk mitigations justify lower estimates, and what ground rules and assumptions diverge. The goal isn't to beat CAPE. It's to narrow the gap through credible technical justification.

ADVANCED

SHOULD-COST VS. WILL-COST: THE SENIOR PM'S MOST POWERFUL TOOL

Should-Cost analysis is a government-led deep dive into contractor cost structure to identify efficiencies. Not just to project what costs will be, but to drive them down. DoDI 5000.02 requires Should-Cost management on major programs. In practice, should-cost teams examine contractor labor mix (are they using senior engineers where journeymen would do?), overhead allocation (is G&A allocation methodology favorable to the program?), and make-buy decisions. The results feed should-cost targets. Negotiating positions for the next contract award. Senior PMs who run aggressive should-cost programs routinely achieve 10-15% cost reductions. The political challenge: your contractor will resist deeply, and your own program office staff (who work with the contractor daily) will resist disrupting the relationship. Push anyway.

Key Terms

ICE

Independent Cost Estimate. An estimate prepared independently of the program office, required at key milestones for ACAT I programs.

CAPE

Cost Assessment and Program Evaluation. OSD office responsible for independent cost analysis of major defense acquisition programs.

Analogous Estimating

Using costs from similar past programs to estimate the new program. Fast but least accurate ($\pm 50\%$). Best for pre-Milestone A.

Parametric Estimating

Using statistical Cost Estimating Relationships (CERs) to relate cost to technical parameters (weight, power, throughput). Moderate accuracy ($\pm 25\%$).

Bottoms-Up Estimating

Estimating each work package or task individually and rolling up to a total. Most accurate ($\pm 10\text{-}15\%$) but requires mature design. Used at Milestone B+.

Cost Risk

The uncertainty or variability in a cost estimate; typically quantified at the 80th percentile (P80) for DoD program budgeting.

LCAT

Labor Category. A defined skill level and discipline (e.g., Software Engineer Level III, Program Manager Level II) that establishes what work a person performs and at what rate they are billed.

Pre-Priced Labor

Labor categories with negotiated, fixed billing rates established in the base contract (e.g., an IDIQ or GWAC), allowing task orders to be placed without re-pricing labor for each award.

Compensation Ratio (Compa-Ratio)

An employee's actual salary divided by the midpoint of their pay band. A ratio of 1.0 means the employee is paid exactly at midpoint. Used to assess whether a contractor's labor pricing is fair and competitive.

Job Code

A contractor-internal identifier linking an employee to a specific LCAT and pay band within their compensation structure. Audited by DCAA to verify labor charging accuracy.

Direct Labor Rate

The base hourly wage paid to an employee before fringe benefits or overhead are applied. The foundation of any labor cost estimate: $\text{Direct Labor Rate} \times \text{Hours} = \text{Direct Labor Cost}$.

Fully Burdened Rate

The total cost per labor hour including all fringe benefits, overhead, G&A, and fee. What the government actually pays per hour of contractor labor.

Knowledge Check

Which cost estimating method is most accurate but requires the most mature program design?

- A. Analogous (top-down)
- B. Parametric

✓ C. Engineering Build-Up (bottoms-up)

- D. Expert judgment

WHY

Engineering Build-Up (Bottoms-Up) estimating, which estimates each individual work package and rolls it up, is the most accurate method ($\pm 10\text{-}15\%$) but requires a mature design and detailed work breakdown structure. It is typically used at or after Milestone B.

A Nunn-McCurdy "critical breach" is triggered when program unit cost growth exceeds what threshold?

- A. 10%
- B. 15%

✓ C. 25%

- D. 50%

WHY

A 25% cost growth above the original baseline triggers a Nunn-McCurdy critical breach, requiring certification by USD(A&S) and Congressional notification. The 15% threshold triggers a "significant breach" requiring notification but not the full certification process.

CAPE's independent cost estimates for major programs are notable because:

A. They are always lower than program office estimates

✓ **B. They are typically higher than program office estimates and historically more accurate**

C. They are only used for ACAT III programs

D. They replace the program office estimate at Milestone B

WHY

CAPE estimates are consistently higher than program office estimates and are statistically more accurate: reflecting the well-documented "optimism bias" in program offices. When CAPE's estimate diverges significantly, it flags the program for additional scrutiny.

The P80 cost estimate represents:

A. 80% of the program's total budget

✓ **B. The cost at which there is an 80% probability that actual costs will be at or below that figure**

C. The 80th percentile of contractor bids received

D. A cost growth of 80% over the baseline estimate

WHY

P80 is the 80th percentile of the cost probability distribution: there is an 80% chance actual costs will be at or below this number. DoD typically uses P80 for budgeting ACAT I programs to provide a reasonable level of confidence that funding will be sufficient.

Which estimating method is most appropriate for an early-phase analysis where the program is not yet well-defined?

A. Engineering Build-Up

B. Parametric

✓ **C. Analogous**

D. Activity-Based Costing

WHY

Analogous estimating: using costs from similar historical programs: is most appropriate in early phases when design details are lacking. While accuracy is limited ($\pm 50\%$), it provides a useful order-of-magnitude estimate to support early programming decisions.

An ICE (Independent Cost Estimate) is required at which program milestone events?

- A. Only at Milestone A
- B. Only at Milestone C

✓ C. At each major milestone decision (A, B, C) for ACAT I programs

- D. Only when a Nunn-McCurdy breach has occurred

WHY

For ACAT I programs, an ICE is required at each major milestone (A, B, C) decision. The ICE is prepared independently from the program office estimate, typically by CAPE at the OSD level or by the Service Cost Center at the component level.

Parametric cost estimating uses what primary analytical tool to relate cost to technical characteristics?

- A. Work Breakdown Structures (WBS)

✓ B. Cost Estimating Relationships (CERs)

- C. Performance Measurement Baselines
- D. Bill of Materials (BOM)

WHY

Parametric estimating relies on Cost Estimating Relationships (CERs): statistical equations derived from historical data that relate cost to technical parameters (e.g., weight, power, throughput). CERs are typically developed from databases of completed programs and are validated against historical actuals.

Historical DoD program data shows that programs overrun their Milestone B estimates by approximately:

- A. 5-10%
- B. 10-15%

✓ C. 20-30%

- D. 50-60%

WHY

DoD's own historical analysis shows programs overrun their Milestone B cost estimates by an average of 20-30%. This chronic "optimism bias" is why CAPE uses higher-confidence (P80) estimates and why Congress enacted Nunn-McCurdy to flag significant overruns.

A Cost Estimating Relationship (CER) would be most useful for which type of cost analysis?

A. Validating a detailed contractor proposal after contract award

✓ **B. Estimating the cost of a new radar system using weight and frequency specifications as independent variables**

C. Calculating earned value variances on an active contract

D. Preparing year-end financial closeout documentation

WHY

CERs are ideal for parametric estimation of new systems where cost can be related to measurable technical parameters. For a radar system, parameters like antenna size, peak power, and frequency range have well-established statistical relationships to cost that can drive early estimates.

When a Nunn-McCurdy critical breach (25%) is declared, the primary consequence if USD(A&S) does not certify the program is:

A. A 1-year program freeze

B. Mandatory program restructuring with a new baseline

✓ **C. Program termination**

D. Transfer to a different ACAT category

WHY

A Nunn-McCurdy critical breach requires certification by the USD(A&S): the program must be re-validated as essential to national security, with reasonable cost and schedule. If USD(A&S) cannot certify the program, it must be terminated. This statutory requirement gives Congress significant leverage over poorly performing programs.

On a pre-priced IDIQ contract, a contractor submits a task order proposal with labor rates 15% above the IDIQ contract ceiling rates. What is the correct course of action?

A. Accept the proposal since ceiling rates are guidelines, not hard limits|||Incorrect. IDIQ ceiling rates are contractually binding, they represent the maximum billable rate for each labor category. A contractor proposing rates above the established ceiling is non-compliant.

✓ B. The proposal is non-compliant: ceiling rates are contractually binding and the contractor cannot exceed them without a contract modification|||Correct. IDIQ labor category ceiling rates establish the maximum rate the contractor may bill. Proposing above the ceiling rate for a task order is a material non-compliance. The contractor must either reduce rates to the ceiling or request a formal contract modification.

C. Request a waiver from DCAA before accepting|||Incorrect. DCAA does not grant waivers for non-compliant proposals. DCAA audits cost proposals and provides opinions to the CO. Waivers for non-compliance would be a CO decision, not a DCAA function.

D. The contracting officer may accept if the work is highly specialized|||Incorrect. Specialization does not override contractual ceiling rates. If work requires skills beyond established LCATs, the solution is to add a new LCAT via contract modification, not to waive the ceiling rate.

WHY

Pre-priced ceiling rates on IDIQ contracts and GWACs are contractually binding. A contractor cannot propose labor rates above the ceiling without a modification to the base contract. A task order proposal that exceeds ceiling rates is non-compliant and must be returned for correction. This is one of the primary government protections in IDIQ vehicles: it prevents rate inflation on individual task orders.

Match each LCAT/labor concept to its correct definition.

CORRECT MATCHES

Compa-Ratio	Employee salary ÷ pay band midpoint; measures pay fairness vs. market
Job Code	Internal HR identifier linking an employee to a specific LCAT and pay band; DCAA audit target
Pre-Priced Labor	Labor categories with rates fixed in the base contract; task orders cannot exceed these rates
Direct Labor Rate	Base hourly wage before fringe, overhead, or G&A are applied

WHY

These four concepts form the backbone of defense contractor labor pricing. DCAA uses job codes and compa-ratios to audit labor charging accuracy. Government PMs use pre-priced rates and direct labor rates to evaluate proposal realism and prevent cost growth.

A contractor's cost proposal for a software sustainment task shows 80% of hours at 'Software Engineer Level III' rates (\$105/hr direct). You know similar tasks at other contractors typically run 60% at Level I/II (\$65-75/hr). What is the most likely issue?

A. The contractor is using an incorrect estimating methodology|||Incorrect for LCAT over-leveling. Over-leveling is a pricing integrity concern where the contractor assigns work to higher-level labor categories than the work requires, driving up cost without improving performance.

✓ B. The contractor is proposing an inflated labor mix: over-leveling LCATs to drive up cost without improving performance|||Correct. Over-leveling (or 'labor-loading') involves proposing Senior Engineer-level hours for work that could be performed by Mid-level or Junior staff. On a T&M or cost-plus contract, this directly inflates billable cost. The CO and DCAA scrutinize the labor mix during cost or price analysis.

C. The direct labor rates are too low for Level III engineers|||Incorrect for over-leveling. Low rates would benefit the government, that would be under-pricing, not over-leveling. The over-leveling problem is the opposite: rates are appropriate for the category, but the wrong (too senior) category is being proposed for the work.

D. The proposal violates TINA because it uses parametric estimating|||Incorrect. TINA violations involve submitting defective cost or pricing data, not using parametric methods. Parametric estimating is a legitimate and commonly used technique.

WHY

'Over-leveling': proposing too many senior LCAT hours on work that could be done by more junior (and less expensive) personnel: is one of the most common ways cost proposals are inflated in defense contracting. As a PM, comparing the proposed labor mix against similar tasks or industry benchmarks is a critical cost realism check. You can challenge this during negotiations by requesting a staffing rationale or comparison to the contractor's actual workforce supporting similar work.

DEFENSE FINANCE & BUDGETING · LESSON 4

Earned Value Management (EVM)

18 min

Learn how to use EVM to measure cost and schedule performance on DoD contracts.

The Dashboard That Tells You the Whole Truth

Imagine you're flying a plane with two gauges: one shows how much fuel you've burned, and one shows how far you've traveled. Separately, each gauge tells you something. Together, they tell you whether you're going to make it to your destination. EVM: Earned Value Management: is both gauges for a defense program. It tells you whether you're getting the work done you planned to get done, at the cost you planned to spend, at any point in time. Without it, a program can look on budget but be six months behind: or look on schedule while burning through money at twice the expected rate. The DoD requires EVM on most contracts over \$50M because "how much have we spent?" only answers half the question. (Raised from \$20M under DFARS Class Deviation 2026-O0011, part of the 2026 DFARS overhaul: verify current DFARS 234.201 before citing the old figure.)

EVM Core Metrics at a Glance

EAC Quick Reference

The Three Numbers, in Plain English

Everything in EVM comes from three numbers measured at the same date. Planned Value is what you planned to have spent by today, according to the baseline. Earned Value is the budgeted value of the work you have actually finished by today. Actual Cost is what you really spent by today. Compare Earned to Planned and you know whether you are ahead or behind schedule. Compare Earned to Actual and you know whether the work is costing more or less than it should. That is the whole system. The dozens of acronyms in the Data module are variations on those two comparisons.

A Worked Example: A \$10M Program at Month Six

A twelve-month, \$10M program planned to be half done by month six, so Planned Value is \$5M. The team has actually completed work worth \$4M of budget, so Earned Value is \$4M. They have spent \$4.5M getting there, so Actual Cost is \$4.5M. Schedule: Earned \$4M against Planned \$5M. The program has done 80% of the work it should have by now. SPI is 0.80. Cost: Earned \$4M against Actual \$4.5M. Every dollar of work is costing \$1.125. CPI is 0.89. If that cost efficiency continues, the \$10M budget will finish at about \$11.25M. That last number is the EAC, and it is the one leadership asks for first.

The Four Formulas You Will Actually Use

$CPI = EV \div AC$ (below 1.0: over cost)

$SPI = EV \div PV$ (below 1.0: behind schedule)

$EAC = BAC \div CPI$ (the simplest forecast)

$TCPI = (BAC - EV) \div (BAC - AC)$ (efficiency needed to finish on budget)

In the example: CPI 0.89, SPI 0.80, EAC \$11.25M, TCPI $(\$10M - \$4M) \div (\$10M - \$4.5M) = 1.09$. The team would need to perform 9% better than plan for the rest of the program to finish on budget, after performing 11% worse so far. That gap is the conversation.

The Example at a Glance

0.89

CPI

Each \$1 of work costs
\$1.125

0.80

SPI

80% of planned work done

\$11.25M

EAC

$BAC \div CPI$

1.09

TCPI

Needed efficiency to hit
BAC

THE 20% THRESHOLD RULE

Research by David Christensen (1993) showed that the CPI at 20% program completion is highly predictive of final CPI. Programs rarely recover a CPI worse than 0.8. If your program shows a CPI of 0.75 at 20% completion, plan for overruns: the data is telling you something systemic is wrong.

EVM Performance Indicator Benchmarks

IPMR REPORTING

The Integrated Program Management Report (IPMR) replaced the CPR and CFSR. The IPMR has seven formats: Format 1 (WBS-based cost/schedule), Format 2 (Organizational), Format 3 (Baseline), Format 4 (Staffing), Format 5 (Problem Analysis), Format 6 (Milestone/IMS), and Format 7 (Explanations). Learn all seven: you'll review these monthly on every major program.

INTERMEDIATE

Using EVM Data to Drive Contractor Behavior

EVM data is only valuable if the government PM uses it to hold contractors accountable. Mid-career PMs should establish a monthly rhythm: review IPMR data within 5 days of receipt, convene a PMR within 10 days, and issue formal corrective action requests for threshold breaches within 15 days. The most powerful

question you can ask a contractor at a PMR: "Your Format 5 says you'll recover 15 schedule days by end of quarter. What specific tasks will be completed early and by whom?" Vague recovery plans are not plans. Require specificity: named individuals, milestone dates, budget planned for the recovery effort. Document these commitments in meeting minutes so the contractor can't walk them back.

ADVANCED

EVM COMPLIANCE SURVEILLANCE AND EVMS ACCEPTANCE: WHAT SENIOR PMS MUST KNOW

DFARS 234.201 requires EVMS on cost or incentive contracts valued at \$50M or more (raised from \$20M in Feb 2026 under DFARS Class Deviation 2026-O0011, part of the DFARS overhaul); a formally validated, government-accepted EVMS is required at \$100M or more (raised from \$50M). But having EVMS is not enough. The system must be compliant with ANSI/EIA-748. DCMA conducts EVMS surveillance and can issue non-compliance findings. A Level III finding (systemic noncompliance) can result in a compliance improvement plan, increased surveillance, and in extreme cases, withholding of fee. Senior PMs must understand their contractor's EVMS compliance status before accepting performance data. An EVMS system that rubber-baselines (retroactively adjusts PMB to reduce variances) produces data that is technically compliant but strategically meaningless. When CPI is suspiciously stable at 1.00 month after month on a complex development contract, question the system. Not just the data.

INTERMEDIATE

Schedule Variance Lies Near the End

SPI has a known flaw: it always drifts back to 1.0 as a program finishes, because eventually all the planned work gets earned, no matter how late. A program that is six months behind will show an SPI of 1.0 on its final day. So SPI is useful early and in the middle, and nearly useless in the last quarter of a program. Experienced PMs pair it with the Integrated Master Schedule: critical path float, milestone dates, and schedule margin tell the truth when SPI has stopped. If a contractor briefs SPI as evidence of schedule health late in a program, ask for the critical path instead.

Management Reserve vs. Undistributed Budget

Two pieces of budget sit outside the work packages and get confused constantly. Management Reserve is budget the contractor holds back from the baseline for unknown in-scope risks. It is not part of the Performance Measurement Baseline, it is not assigned to any work, and using it requires a documented change to the baseline. Undistributed Budget is budget for authorized work that has not yet been planned into work packages, often because a mod was just received. It is part of the contract budget base but not yet in the PMB. A contractor that keeps burning MR to cover routine overruns is masking a CPI problem. A

large UB balance months after a mod means planning is behind. Both show up in the IPMR and both are worth a question at every review.

Budget Buckets in an EVMS

BUCKET	IN THE PMB?	WHAT IT IS FOR	WARNING SIGN
Work packages	Yes	Planned, scheduled, budgeted work	Frequent replanning of open packages
Planning packages	Yes	Future work not yet detailed	Large balance close to the current period
Undistributed Budget	No (in CBB)	Authorized scope awaiting planning	Sitting unplanned months after a mod
Management Reserve	No (in CBB)	Contractor's reserve for in-scope risk	Steady drawdown to cover overruns
Profit / fee	No	Above the CBB entirely	n/a

ADVANCED

The Integrated Baseline Review: What to Look For

The IBR, held within months of award, is the government's chance to confirm the contractor's baseline is realistic before the EVM data starts flowing. Senior PMs use it to test four things. Is the scope in the work packages the same as the scope in the SOW, with nothing missing and nothing extra? Is the schedule logic real, with predecessors, successors, and a critical path that survives a challenge? Is the budget time-phased the way the work will actually happen, or front-loaded to make early performance look good? And is Management Reserve sized to the program's risk register rather than to a percentage? A weak IBR produces a baseline that generates comfortable data for a year and then collapses. A demanding IBR is the cheapest risk mitigation on the program.

Rebaselining: Over Target Baselines and When They Are Legitimate

When a program's baseline becomes unachievable, the contractor may request an Over Target Baseline (more budget than the contract budget base) or an Over Target Schedule (dates beyond the contract schedule). Done properly, an OTB acknowledges the real cost to complete so that future EVM data is meaningful again. Done badly, it erases variances and resets CPI to 1.0 with no change in performance, which is the "rubber baseline" problem. The government must approve an OTB. Senior PMs approve one only when the remaining work is replanned bottom-up, variances to date are preserved in the record, and the new baseline is tied to a realistic EAC. A request to rebaseline without those three is a request to stop measuring.

READING BETWEEN THE FORMATS

The Data module's IPMR lesson covers the report formats. The senior PM habit is cross-checking them: Format 1 variances against Format 5 explanations (does every big variance have a real narrative?), Format 3 baseline changes against MR and UB logs (where did the budget go?), and Format 6 milestone dates against the SPI story (does the schedule agree with the index?). Inconsistencies between formats are where the truth usually is.

Key Terms

EVM (Earned Value Management)

The system that tells you whether a program is getting the work done it planned to get done, at the cost it planned to spend. Think of it as the program's health dashboard. Not just 'how much have we spent' but 'are we getting our money's worth?'

BCWS (PV)

Budgeted Cost of Work Scheduled (Planned Value). What you planned to spend by now.

BCWP (EV)

Budgeted Cost of Work Performed (Earned Value). The budget value of work actually accomplished.

ACWP (AC)

Actual Cost of Work Performed (Actual Cost). What you actually spent.

CPI

Cost Performance Index. Efficiency of cost performance (EV/AC).

SPI

Schedule Performance Index. Efficiency of schedule performance (EV/PV).

EAC

Estimate at Completion. The projected total cost to complete the program.

Knowledge Check

A program has: PV = \$10M, EV = \$8M, AC = \$9M. What is the Cost Performance Index (CPI)?

A. 1.125

✓ B. 0.89

C. 0.80

D. 1.25

WHY

$CPI = EV / AC = \$8M / \$9M = 0.89$. This means for every dollar spent, the program is only earning \$0.89 of planned value: the program is over budget. The SPI would be 0.80 ($EV/PV = \$8M/\$10M$), meaning it's also behind schedule.

Using the most statistically accurate EAC formula for a program that is 30% complete, BAC = \$100M and CPI = 0.85:

A. \$100M

B. \$115M

✓ C. \$117.6M

D. \$85M

WHY

$EAC = BAC / CPI = \$100M / 0.85 = \$117.6M$. This formula is the most accurate predictor once a program is more than 20% complete. The projected overrun of \$17.6M signals a systemic cost efficiency problem.

What does a Schedule Performance Index (SPI) of 0.80 indicate?

A. The program is 20% over budget

✓ B. For every \$1.00 of work planned, only \$0.80 of work has been accomplished: behind schedule

C. The program has completed 80% of its total work

D. The schedule has slipped by 20 days

WHY

$SPI = EV / PV = 0.80$ means only 80 cents of planned work has been accomplished for every dollar of work scheduled. The program is behind schedule. $SPI < 1.0$ always means behind schedule; $SPI > 1.0$ means ahead of schedule.

The To-Complete Performance Index (TCPI) represents:

- A. The CPI achieved so far on the program
- ✓ **B. The cost efficiency required on remaining work to achieve the EAC**
- C. The ratio of planned work to actual work
- D. The schedule efficiency required to meet the deadline

WHY

$TCPI = (BAC - EV) / (EAC - AC)$. It tells you the cost efficiency you must achieve on all remaining work to hit your EAC. A TCPI > 1.10 is generally considered unrealistic: it means you need to be significantly more efficient than you've been, which rarely happens.

EVM is contractually required on DoD contracts above what minimum threshold?

- A. \$5M
- B. \$20M
- ✓ **C. \$50M**
- D. \$100M

WHY

DFARS 234.201 requires Earned Value Management System (EVMS) compliance on DoD contracts above \$50M (raised from \$20M in February 2026 under DFARS Class Deviation 2026-O0011, part of the DFARS overhaul). Programs above \$100M (raised from \$50M) require a formal, government-validated EVMS that meets ANSI/EIA-748 criteria.

In EVM, the Performance Measurement Baseline (PMB) is:

- A. The original program budget as approved at Milestone B
- ✓ **B. The time-phased budget plan against which actual performance is measured (BAC minus MR)**
- C. The contractor's cost estimate for the remaining work
- D. The government's independent cost estimate

WHY

The PMB is the time-phased budget baseline for all authorized work, equal to BAC minus Management Reserve (MR). MR is held above the PMB by the PM and not included in the performance measurement baseline. Variances are measured against the PMB, not the total program budget.

A Cost Variance (CV) of -\$2M means:

- A. The program is \$2M ahead of schedule
- B. The program has spent \$2M less than planned

✓ C. The program has spent \$2M more than the earned value of work accomplished

- D. The program will overrun by \$2M at completion

WHY

CV = EV - AC. A negative CV (-\$2M) means the program has spent \$2M more (AC) than the budget value of work accomplished (EV). This is a cost overrun on the work performed to date. It does NOT directly state the final overrun: that requires an EAC calculation.

Which IPMR format specifically covers problem analysis: explaining the root cause of significant variances?

- A. Format 1
- B. Format 3

✓ C. Format 5

- D. Format 7

WHY

IPMR Format 5 is the "Problem Analysis Report": it requires the contractor to explain significant variances (typically CV% or SV% above threshold), identify root causes, and describe corrective action plans. This format is often the most closely scrutinized by program managers.

According to Christensen's research, when a program's CPI is established at what completion percentage, it rarely improves significantly?

- A. 10%
- B. 15%

✓ C. 20%

- D. 50%

WHY

Christensen's landmark 1993 study found that the CPI at 20% program completion is a highly reliable predictor of final CPI, and that the final CPI is almost always worse than the CPI at 20% completion. This is why early EVM analysis is critical: problems caught at 15% are far cheaper to fix than at 50%.

Management Reserve (MR) in an EVMS context is:

- A. Budget included in the PMB for identified risks
- ✓ **B. Budget held outside the PMB by the PM to handle unplanned work or risks**
- C. The contractor's profit margin on a cost-plus contract
- D. Undistributed budget awaiting work package assignment

WHY

Management Reserve is budget held by the PM above (outside) the Performance Measurement Baseline. It is not in the PMB, not distributed to work packages, and requires formal authorization to use. MR covers unforeseen in-scope work. Undistributed Budget (UB) is different: it's budget not yet assigned to specific work packages but IS within the PMB.

At month six, PV = \$5M, EV = \$4M, AC = \$4.5M, BAC = \$10M. What is the EAC using $BAC \div CPI$?

- A. \$10M
- ✓ **B. \$11.25M**
- C. \$12.5M
- D. \$9M

WHY

$CPI = \$4M \div \$4.5M = 0.89$. $EAC = \$10M \div 0.89 \approx \$11.25M$.

Why is SPI unreliable in the final months of a program?

- A. Because costs are all incurred
- ✓ **B. Because SPI always converges to 1.0 as all planned work is eventually earned, regardless of how late**
- C. Because DCMA stops collecting it
- D. Because Management Reserve is exhausted

WHY

Late in a program, use the IMS critical path and milestone dates instead of SPI.

A contractor requests an Over Target Baseline. Which condition makes it legitimate rather than a "rubber baseline"?

A. CPI is reset to 1.0 with no other change

✓ B. Remaining work is replanned bottom-up, variances to date are preserved, and the new baseline ties to a realistic EAC

C. The request is made in the last month of the contract

D. Management Reserve is increased to cover past overruns

WHY

An OTB is meant to restore meaningful measurement. Erasing variances without replanning defeats the purpose.

Match the budget bucket to its description:

CORRECT MATCHES

Management Reserve	Held outside the PMB for unknown in-scope risk
Undistributed Budget	Authorized scope not yet planned into work packages
Work package	Planned, scheduled, budgeted work inside the PMB

WHY

MR is held outside the PMB for in-scope risk. UB is authorized scope not yet planned. Work packages are planned, budgeted work inside the PMB.

DEFENSE FINANCE & BUDGETING · LESSON 5

Estimate at Completion (EAC)

14 min

Learn what EAC and ETC actually mean in plain English, and why a contract ceiling is not the same as the money that is actually funded and available to spend.

What EAC Actually Is

EAC stands for Estimate at Completion. In plain terms, it is your best current guess at what the whole program will cost by the time it is done. Not what you planned to spend (that is the BAC). What you now think you will actually spend, based on how the work has really gone so far. If you have spent more than planned for the work you have completed, your EAC climbs above your BAC. If you have somehow come in ahead, it can drop below. Either way, EAC is the forecast that tells everyone above you whether this program is still on track or heading for trouble.

A Concrete Example: Building One Person's Number

Let's make this real with one person instead of a formula. Say John does cybersecurity work on a services contract in Europe. His direct labor rate is \$80 an hour, and his wrap rate is 1.7, so his fully burdened labor cost comes out to $\$80 \times 1.7 = \136 an hour. If he is scheduled for around 1,920 hours this year (a standard work year once you account for holidays and leave), his labor alone costs the program $\$136 \times 1,920 = \$261,120$. Now add his Other Direct Costs (ODCs). Because he is working overseas, his contract authorizes a Living Quarters Allowance (LQA) of \$40,000 a year to help cover housing, and a Cost of Living Allowance (COLA) of \$10,000 a year to cover the higher cost of living in that location. That is \$50,000 in direct ODCs. Those ODCs still pick up the company's G&A rate, say 10% here, which adds another \$5,000, bringing his total ODC cost to \$55,000. Add it up: \$261,120 in labor plus \$55,000 in ODCs comes to about \$316,000 for John alone, for one year. Now picture doing that same math for every person on the contract, plus materials, plus travel, plus every other ODC, and rolling it all up into one number. That rolled-up total, refreshed every month with real hours and real costs instead of the original plan, is what an EAC actually is. It is not some number finance pulls out of thin air. It is this kind of buildup, done at scale, updated with reality.

ETC: What's Left to Spend

EAC tells you the total. ETC tells you what is still left to spend to get there. The formula is simple: $ETC = EAC - AC$ (Actual Cost, what has actually been spent so far). Picture John again. Say it is mid-year and the program has already spent \$150,000 of his labor and ODCs, that is his AC. His annual EAC is still \$316,120. So his ETC, the money still needed to finish the year, is $\$316,120 - \$150,000$, which is \$166,120. AC tells you where you have been. EAC tells you where you are headed. ETC tells you how

much runway you actually have left, and that last number is the one that matters most for the next topic: whether the money to cover it is actually available.

Contract Ceiling Is Not the Same as Funded Value

Here is a mistake that trips up a lot of new PMs. A contract or task order having a \$2,000,000 ceiling does not mean \$2,000,000 is available to spend today. The ceiling is just the maximum the contract could ever be worth. Funded Value is different: it is the amount of money the government has actually obligated to that contract right now, through a specific funding document tied to an ACRN (Accounting Classification Reference Number). A contractor can only legally bill against the Funded Value, never the ceiling. On a lot of DoD services and R&D work, funding shows up incrementally instead of all at once, because O&M and RDT&E dollars are annual or two-year money tied to the budget cycle. A three-year task order with a \$2,000,000 ceiling might only have \$600,000 funded in year one, with more added later as Congress appropriates it and the government issues new funding modifications.

LIMITATION OF FUNDS: THE CLAUSE THAT MAKES THIS REAL

Take the numbers above: \$600,000 funded, \$450,000 spent, so \$150,000 of funded balance remains. If the ETC to finish the current scope is \$210,000, a \$60,000 shortfall is coming. This exact scenario is why FAR 52.232-22 (Limitation of Funds) and FAR 52.232-20 (Limitation of Cost) exist, and it is one of the most important clauses you will ever apply as a PM. On an incrementally funded cost-reimbursement contract, the contractor must notify the government in writing once they project hitting a set percentage of the funded amount, commonly 75%, with enough lead time for the government to act. From there, the government either processes a funding modification to add more money, or the contractor stops work at the funded ceiling. A contractor is not obligated to keep working past their funded amount, and a government PM who pressures them to do so anyway is creating an Anti-Deficiency Act problem, not just an awkward conversation.

Why This Number Keeps PMs Up at Night

I have sat in program reviews where the entire room goes quiet the moment EAC comes up. Here is why: unlike the BAC, which is just the plan you started with, EAC reflects reality, what you have actually spent and what you have actually gotten done. Senior leaders use it to decide whether your program keeps going, gets restructured, or needs a hard conversation about scope. On the largest weapon system programs, a big enough EAC overrun can trigger a formal Nunn-McCurdy breach, but that is scoped to major defense acquisition programs, not something every PM needs to track. What every PM does need to do is report their EAC honestly, every single period. It is not paperwork. It is your credibility on the line.

EAC MATTERS FAR LESS ON A FIRM-FIXED-PRICE CONTRACT

Everything above assumes a cost-reimbursement contract (CPFF, CPIF, CPAF), where the government pays actual costs and needs to track where those costs are headed. On a Firm-Fixed-Price (FFP) contract, the price was locked at award. If the contractor's EAC blows past their internal budget, that is their profit disappearing, not government money. You still watch schedule and technical performance on FFP, just not the contractor's internal EAC.

Quick Gut-Check Before You Report an EAC

- Do you know your ETC (EAC minus AC), not just the total EAC?
- Do you know your current Funded Value, not just the contract ceiling?
- Is your ETC covered by the Funded Value that remains, or is a shortfall coming?
- If a shortfall is coming, has the 75% notification already gone out, well before you hit the wall?

THE GOLDEN RULE OF EAC

Never manage to the EAC you want. Manage to the EAC the data is telling you. PMs who quietly sandbag their number to avoid scrutiny always get caught eventually, and it costs them far more credibility than an early, uncomfortable, honest number ever would.

Key Terms

EAC

Estimate at Completion. Your current best forecast of what the entire program will cost when it is finished, built from real performance data instead of the original plan.

ETC

Estimate to Complete. The projected cost of the work still left to do: $ETC = EAC - AC$.

AC

Actual Cost. What has actually been spent on the program or task order so far.

BAC

Budget at Completion. The total approved budget for all authorized work on the program (the baseline you started with).

Contract Ceiling

The maximum dollar amount a contract or task order could ever be worth. This is a cap, not real money sitting in an account.

Funded Value

The amount of money actually obligated to a contract right now, tied to a specific funding document. This is the real money a contractor can legally bill against today, separate from the ceiling.

ACRN

Accounting Classification Reference Number. The specific funding line on a contract that ties obligated dollars to a particular appropriation and CLIN.

Limitation of Funds / Limitation of Cost

FAR 52.232-22 and 52.232-20. The clauses governing incrementally funded cost-reimbursement contracts, requiring the contractor to notify the government (commonly at 75% of funded value) before they run out of money.

Knowledge Check

What does EAC (Estimate at Completion) actually represent?

A. The original approved budget for the program

✓ **B. Your current best forecast of what the entire program will cost when finished, based on real performance**

C. The amount of money currently obligated to the contract

D. The contractor's profit margin at contract completion

WHY

EAC is a living forecast built from actual cost and performance data, not the static original plan (that is the BAC). It updates as work actually happens.

ETC (Estimate to Complete) is calculated as:

A. $EAC + AC$

✓ **B. $EAC - AC$**

C. $BAC - AC$

D. BAC / EAC

WHY

$ETC = EAC - AC$. It tells you the cost of the work still left to do, separate from what has already been spent (AC) and the total forecast (EAC).

A program has an EAC of \$316,120 and has spent \$150,000 to date (AC). What is the ETC?

✓ **A. \$166,120**

B. \$316,120

C. \$466,120

D. \$150,000

WHY

$ETC = EAC - AC = \$316,120 - \$150,000 = \$166,120$. That is the money still needed to finish the remaining work.

What is the key difference between a contract's ceiling and its Funded Value?

A. They are always the same number

✓ **B. Ceiling is the maximum the contract could ever be worth; Funded Value is the amount actually obligated and available to bill against right now**

C. Funded Value is always higher than the ceiling

D. Ceiling only applies to FFP contracts

WHY

The ceiling is a cap on what the contract could grow to. Funded Value is the actual obligated dollars tied to a funding document (via an ACRN) that the contractor can legally bill against today. A contract can have a large ceiling and a much smaller Funded Value.

Why do many DoD services and R&D contracts get funded incrementally instead of all at once?

A. Contracting officers prefer smaller numbers

✓ **B. O&M and RDT&E appropriations are annual or multi-year money tied to the budget cycle, so funding often arrives in increments as it is appropriated**

C. Incremental funding is required on all FFP contracts

D. It reduces the contractor's fee

WHY

Because O&M (1-year) and RDT&E (2-year) money follows the PPBE and appropriations cycle, a multi-year effort often only has the current increment funded, with more added later as Congress appropriates and the government issues funding mods.

An ACRN (Accounting Classification Reference Number) is best described as:

A. The contractor's internal cost code

✓ **B. The specific funding line on a contract that ties obligated dollars to a particular appropriation**

C. A type of contract vehicle

D. The government's cost estimate for a task order

WHY

ACRNs tie specific dollar amounts and appropriation types to specific CLINs. They are how the government tracks exactly which funding covers which piece of work.

A contract has \$600,000 Funded Value, \$450,000 spent (AC), and an ETC of \$210,000 to finish current scope. What does this tell you?

A. Everything is fine, the ceiling has plenty of room

✓ B. There is a \$60,000 funding shortfall: the remaining funded balance (\$150,000) will not cover the ETC (\$210,000)

C. The contractor has been overpaid

D. The EAC must be recalculated using CPI

WHY

Remaining funded balance = \$600,000 - \$450,000 = \$150,000. ETC is \$210,000. That is a \$60,000 shortfall against the funded amount, regardless of how much room is left under the ceiling.

What does FAR 52.232-22 (Limitation of Funds) require a contractor to do?

A. Stop billing the government entirely once awarded

✓ B. Notify the government in writing once they project reaching a set percentage (commonly 75%) of the funded amount, with enough lead time for the government to act

C. Automatically increase the contract ceiling

D. Refund unspent funds at year end

WHY

Limitation of Funds (incrementally funded cost-reimbursement contracts) requires early written notification, typically at the 75% threshold, so the government has time to add funding or the contractor can stop work at the funded limit without a lapse.

If a contractor's funded balance runs out and the government PM pressures them to keep working anyway without a funding mod, what is the real risk?

A. The contractor's insurance premium increases

✓ B. A potential Anti-Deficiency Act violation, since the government would be directing work beyond what has been legally obligated

C. The contract automatically converts to FFP

D. Nothing, as long as the work eventually gets funded

WHY

The government cannot obligate or direct spending beyond what Congress has appropriated and the government has obligated. Directing a contractor to keep working past the funded limit without proper funding is exactly the kind of situation the Anti-Deficiency Act exists to prevent.

Why does EAC matter far less to the government on a Firm-Fixed-Price (FFP) contract?

A. FFP contracts are not allowed to have an EAC

✓ B. The price is locked at award, so if the contractor's EAC exceeds their internal budget, that is the contractor's profit disappearing, not government money

C. The government still pays the contractor's full EAC on FFP contracts

D. FFP contracts never experience cost growth

WHY

On FFP, the contractor bears the cost risk. The government PM should still watch schedule and technical performance, but does not need to scrutinize the contractor's internal EAC the way they would on a cost-reimbursement contract.

A PM who 'manages to the EAC they want' instead of the EAC the data shows is:

A. Following best practice for program stability

✓ **B. Sandbagging, which erodes credibility once leadership eventually sees the real numbers**

C. Required to do this under DFARS

D. Only a problem on contracts above \$50M

WHY

An artificially stable or optimistic EAC that does not reflect real performance is a red flag to experienced leadership. Reporting an accurate, even uncomfortable, number early protects a PM's credibility far more than a comfortable number that later blows up.

DEFENSE FINANCE & BUDGETING · LESSON 6

Wrap Rates & Cost Structure

25 min

Understand how defense contractors build their prices. Wrap rates, fringe, overhead, G&A, fee/profit, and how PMs use this knowledge to evaluate proposals and control costs.

What a Wrap Rate Actually Is

A wrap rate is the total price a contractor charges you for one hour of someone's labor, once every layer of cost gets stacked on top of their actual paycheck. Direct labor, fringe benefits, overhead, G&A, and fee all get wrapped into that single number, which is exactly why it is called a wrap rate. It is almost always two to three times higher than what the employee actually takes home. Most PMs focus entirely on total contract price. But the price is built from dozens of individual cost elements that a contractor controls.

Understanding how a contractor builds their price gives you critical leverage: you can identify inflated overhead, challenge unrealistic rates, spot rate trends that predict cost growth, and negotiate better deals. More importantly, when DCAA audits a contractor's accounting system and finds non-compliant cost pools or unsupported rates, your contract is at risk. Knowing the structure means you can ask the right questions before signing.

MEET SARAH: ONE ENGINEER, FIVE LAYERS OF COST

The easiest way to follow this lesson is to track one real person the whole way through. Sarah is a systems engineer. Her actual paycheck comes out to \$65 an hour, base salary only. That \$65 is her direct labor rate, the first layer. Every other term in this lesson is just another layer stacked on top of that \$65 before the government sees a bill. By the end of this lesson you will watch Sarah's \$65 an hour turn into over \$139 an hour, and you'll know exactly which layer added what and why.

Direct Costs: What the Contractor Directly Charges Your Contract

Direct costs are the foundation of any cost estimate. Direct Labor is Sarah's \$65 an hour, the hours your program pays for directly, billed at the contractor's direct labor rate for her labor category (e.g., Systems Engineer Level III). Direct Materials. Hardware, software licenses, test equipment, and supplies that can be traced to your contract. Other Direct Costs (ODCs). Travel, subcontractor costs, consultant fees, special test equipment. Direct costs are identifiable, auditable, and must be allocable to your contract under CAS and FAR Part 31.

ODCs Up Close: LQA, COLA, M&IE, and Why They Are Not All Treated the Same

Other Direct Costs is a big bucket, so it helps to know what is actually inside it. On a services contract with people working overseas, the three you will run into constantly are LQA (Living Quarters Allowance, a housing allowance), COLA (Cost of Living Allowance, covering a higher cost of living at that location), and M&IE (Meals & Incidental Expenses, the meals portion of per diem for travel or short-term assignments). All three are direct costs tied to a specific person, not overhead the company spreads across everyone. How these get treated depends entirely on your contract type. On a cost-reimbursement contract (CPFF, CPIF, CPAF), the government reimburses these ODCs at actual allowable cost, plus the contractor's G&A rate, as long as they are authorized in the contract or an approved compensation plan. On a Time & Materials (T&M) contract, the hourly labor rates you negotiated do not include ODCs. Travel, LQA, COLA, and M&IE get billed separately, and per FAR 52.232-7, the contractor generally cannot add fee or profit on top of those ODCs, they get billed at cost. On a Firm-Fixed-Price (FFP) contract, none of this gets separately reimbursed. The contractor built their estimate of these costs into the fixed price at proposal time, and from then on, they eat any difference between what they estimated and what it actually costs. Know which contract type you are on before you assume an ODC is automatically getting paid back.

Building a Wrap Rate from the Ground Up

Typical Rate Ranges by Contractor Size (Defense Industry)

Overhead vs. G&A: The Critical Distinction

Overhead and G&A are both indirect costs, but they are structured very differently. Overhead is typically division or department-level. It covers the cost of running Sarah's specific business unit (facilities for her division, her department manager's salary, indirect engineers, IT servers for that group). A company with three divisions will have three separate overhead pools, each with a different rate. G&A is company-wide. It covers the CEO's salary, corporate HR, corporate finance, corporate legal, and business development costs that benefit the entire enterprise, Sarah's division included. G&A is applied to Total Cost Input (direct + overhead) as the final layer before fee. Understanding this two-layer structure is critical for evaluating proposals.

FORWARD PRICING RATE AGREEMENTS (FPRAS): WHAT THEY ARE AND WHY THEY MATTER

An FPRA is a negotiated agreement between a large contractor and DCAA/ACO that establishes the indirect cost rates to be used for pricing future proposals for a set period (typically 1-2 years). Once an FPRA is in place, contractors must use those rates in all proposals. They cannot negotiate rates individually on each contract. For PMs, FPRAs mean two things: (1) you know what rates to expect from major contractors. They are public record in FPRA letters; (2) if a contractor's actual rates come in higher than their FPRA rates, the government may be owed money via an equitable adjustment.

Fee Limits Under DFARS: What Profit Is Actually Capped At

The Weighted Guidelines Method: How Profit is Determined

REAL TALK

Nobody grows up dreaming of learning Cost Accounting Standards, and on most programs you genuinely don't need to. CAS exists so a contractor has to count costs the same way on every job, which is how the government catches someone quietly gaming the numbers. It only kicks in around \$35 million and up, so if that's not your program, this is your company's accounting department's problem, not yours. Read the table once so the term rings a bell, then go back to running your actual job.

Cost Accounting Standards (CAS): What PMs Need to Know

READING A CONTRACTOR'S INCURRED COST SUBMISSION (ICS)

Every year, contractors on cost-reimbursable contracts must submit an Incurred Cost Submission (ICS). Also called an Incurred Cost Proposal or ICP. Showing their actual indirect cost rates for the year. DCAA audits these submissions to verify that final rates match what was billed to contracts. If actual overhead ends up lower than the rates billed to your contract, the contractor owes the government a refund. If actual rates are higher, the government may owe the contractor more. As a PM, you should know your major contractors' ICS filing status. Delayed ICS submissions (often years late) are a compliance red flag.

UNALLOWABLE COSTS: THE FAR PART 31 TRAP

FAR Part 31 defines which costs are allowable on government contracts. Certain costs are expressly unallowable. Meaning the contractor CANNOT bill them to government contracts, even if they are legitimately incurred. Key unallowable costs: entertainment and gifts, lobbying and political contributions, alcoholic beverages, certain advertising costs, costs of organizing or reorganizing the contractor's business, excessive executive compensation above the statutory cap (currently ~\$600K/year for the 5 highest-paid executives), fines and penalties. If DCAA discovers unallowable costs in an incurred cost audit, the contractor must reimburse the government. PMs should flag unusually high G&A rates, which may indicate unallowable costs improperly bundled into cost pools.

INTERMEDIATE

Auditing Contractor Indirect Rates During Performance

Mid-career PMs on cost-type contracts must understand that the rates billed on invoices are provisional. Final rates are not established until DCAA audits and negotiates final billing rates, typically 2-3 years after

the accounting period closes. This creates a reconciliation risk: if final rates are higher than provisional rates, the contractor gets a retroactive billing increase. Track your contractor's provisional vs. forward pricing rate agreements (FPRAs). When provisional rates diverge significantly from actuals, push for an FPRA update. PMs who ignore this find themselves in a final rate reconciliation dispute 3 years after contract closeout. When the contractor is trying to bill \$2M in additional indirect costs that weren't budgeted.

ADVANCED**UNALLOWABLE COSTS, DIRECT CHARGING, AND THE DCAA AUDIT TRAIL**

FAR Part 31 defines cost allowability. What contractors can bill to the government. Unallowable costs (entertainment, lobbying, certain IR&D that doesn't qualify, executive compensation above the cap) must be segregated and excluded from billings. DCAA's job is to verify this segregation. What senior PMs need to know: if DCAA finds unallowable costs billed to your contract, the contractor must repay them with interest. And the PM has an obligation to pursue collection. The more dangerous scenario is "direct charging" problems: a contractor billing costs directly to your contract that should be in indirect pools (inflating the direct cost base artificially). This is fraud when intentional. Look for anomalous direct labor charges. Senior executives billed at 100% direct to a single contract, or subcontractor pass-through markups that exceed the prime's normal margins.

Key Terms

Direct Cost

A cost that can be specifically identified with a single final cost objective (a contract). Examples: direct labor hours, direct materials, travel directly supporting a contract.

Indirect Cost

A cost that cannot be directly attributed to a single contract and must be allocated across multiple cost objectives. Examples: facility rent, HR, finance, IT infrastructure.

Fringe Benefits Rate

The rate applied to direct labor to cover employee benefits. Health insurance, retirement, FICA, paid leave. Typically 25-35% of direct labor.

Overhead Rate

Indirect costs allocated to direct labor or direct costs within a specific organizational unit (division, department). Covers facilities, supervision, equipment, and indirect labor.

G&A Rate

General and Administrative rate. Company-wide indirect costs covering executive management, corporate HR, legal, finance, and business development. Applied to total cost input.

Wrap Rate

The total billing rate for one hour of labor, combining direct labor + fringe + overhead + G&A + fee. The all-in cost of one hour of contractor work.

Fee / Profit

The contractor's profit on a contract. On cost-type contracts, fee is negotiated separately; on fixed-price contracts, profit is embedded in the price. DFARS limits fee rates by contract type.

Cost Pool

A grouping of indirect costs that are accumulated and then allocated using a common allocation base (e.g., a facilities cost pool allocated based on square footage).

Allocation Base

The measure used to distribute indirect costs to contracts (e.g., direct labor hours, direct labor dollars, total cost input).

CAS

Cost Accounting Standards, 19 standards (48 CFR 9900) governing how defense contractors accumulate, measure, and allocate costs. Under the FY2026 NDAA, the per-contract CAS trigger is \$35M and full CAS coverage kicks in at \$100M in annual covered contract awards (both figures replaced the older \$2.5M and \$50M thresholds).

CASB

Cost Accounting Standards Board. The federal board that promulgates CAS. Contractors must disclose their accounting practices in a Disclosure Statement (CASB DS-1).

Forward Pricing Rate Agreement

FPRA. A negotiated agreement between the contractor and the government on indirect cost rates for use in forward pricing of proposals. Eliminates rate negotiation on every proposal.

LQA

Living Quarters Allowance. An allowance authorized for employees working overseas to help cover housing costs. Paid as a direct cost tied to that specific person's assignment.

COLA

Cost of Living Allowance. An allowance authorized for employees working in a higher cost-of-living location overseas, covering the cost differential compared to their home base. Paid as a direct cost tied to that specific person's assignment.

M&IE

Meals & Incidental Expenses. The meals and incidentals portion of a per diem rate for employees traveling or on short-term assignment away from their home base, set using government per diem tables.

Knowledge Check

A contractor's Systems Engineer earns \$70/hr direct labor. Fringe = 33%, Overhead = 55% of DLR, G&A = 13% of total cost input, Fee = 9%. What is the approximate wrap rate?

A. \$70/hr

B. \$148/hr

✓ C. \$168/hr

D. \$192/hr

WHY

Step by step: DLR = \$70. + Fringe (33%): $\$70 \times 0.33 = \$23.10 \rightarrow \$93.10$. + Overhead (55% of DLR): $\$70 \times 0.55 = \$38.50 \rightarrow \$131.60$. + G&A (13% of \$131.60): $\$17.11 \rightarrow \148.71 . + Fee (9%): $\$13.38 \rightarrow \$162.09 \approx \$162/\text{hr}$. Closest to \$168/hr at these rounding conventions. The multiplier is approximately 2.4× DLR. Typical for a mid-size contractor.

Which cost element covers company-wide expenses like executive management, corporate legal, and enterprise IT. Applied as the final indirect rate layer before fee?

A. Fringe Benefits

B. Overhead

✓ C. G&A (General and Administrative)

D. Management Reserve

WHY

G&A covers company-wide administrative costs that benefit the entire enterprise. Not just a single division. It is applied to Total Cost Input (all direct + overhead costs) as the final indirect cost layer before fee. Overhead is division/department-level; G&A is corporate-level.

Under FAR 15.404-4, the maximum fee on a Cost-Plus-Fixed-Fee (CPFF) completion contract is:

A. 5% of estimated cost

B. 7% of estimated cost

✓ C. 10% of estimated cost

D. 15% of estimated cost

WHY

CPFF completion contracts are capped at 10% of estimated cost per FAR 15.404-4(c)(4)(i). CPFF term contracts are capped at 7%. These caps exist because on cost-reimbursable contracts, there is no upper limit on cost. Unlimited fee on unlimited cost would remove all incentive for cost control.

An FPRA (Forward Pricing Rate Agreement) benefits the government primarily by:

A. Allowing contractors to charge any rate they choose per contract

✓ B. Establishing negotiated indirect rates applicable to all future proposals, eliminating per-contract rate negotiation

C. Capping profit at 10% on all contracts

D. Requiring DCAA audit of every individual proposal

WHY

An FPRA establishes pre-negotiated indirect rates (overhead, G&A, fringe) that the contractor must use in all proposals for the agreement period. This saves significant negotiation time, provides rate predictability, and allows the government to evaluate proposals against known rates rather than contractor-asserted rates.

Under CAS 402, a contractor discovers it has been classifying a particular engineer's time as "indirect" on commercial contracts but "direct" on government contracts. This is:

A. Acceptable as long as total costs are the same

✓ B. A CAS 402 violation: costs must be treated consistently as direct or indirect across all contracts

C. Permitted since commercial contracts have different rules

D. Only a problem if DCAA identifies it

WHY

CAS 402 requires consistent treatment of similar costs. A cost cannot be treated as direct on government contracts and indirect on commercial contracts (or vice versa). This is one of the most frequently cited CAS violations in DCAA audits. Inconsistent treatment allows contractors to maximize government billings by allocating their most expensive costs directly to government contracts.

Which of the following is expressly UNALLOWABLE under FAR Part 31 and cannot be billed to government contracts?

A. Direct engineer labor on contract tasks

B. Fringe benefits for employees working on the contract

✓ C. Entertainment costs and gifts

D. Facilities overhead for space used to perform the contract

WHY

FAR 31.205-14 expressly identifies entertainment costs as unallowable. Other key unallowable costs include: lobbying (31.205-22), alcoholic beverages (31.205-51), advertising costs not related to recruitment or product sale (31.205-1), and excessive executive compensation above the statutory cap. When DCAA finds unallowable costs in indirect pools, the government is owed a credit adjustment.

A contractor's wrap rate multiplier is 3.8× their direct labor rate. Compared to an industry average of 3.0×, what should a PM do?

A. Accept it since contractors set their own rates

✓ B. Request an explanation and FPRA documentation; question overhead and G&A rates specifically

C. Immediately terminate the contract

D. Use the rate only if the contract is cost-plus

WHY

A 3.8× multiplier is above typical industry range (2.5-3.5× for most defense contractors). The PM should request the FPRA letter or rate documentation, identify which pool (overhead or G&A) is driving the high rate, and challenge rates that are not supported by audited actuals. High rates may indicate excessive overhead, unallowable costs in pools, or the contractor's rates not being DCAA-audited.

What is the purpose of a contractor's annual Incurred Cost Submission (ICS)?

A. To propose new labor categories for the upcoming year

✓ B. To report actual indirect cost rates for the completed year, enabling reconciliation with billed rates

C. To submit a new FPRA request

D. To report earned value performance data

WHY

The ICS (also called Incurred Cost Proposal) reports a contractor's actual indirect rates for the fiscal year, which are compared to the provisional or FPRA rates that were billed to contracts during the year. If actual rates were lower than billed, the contractor owes a credit. If higher, the government may owe additional costs on cost-reimbursable contracts. DCAA audits ICS submissions. Delays of years are common and represent a significant financial risk to the government.

The Weighted Guidelines Method (DFARS 215.404-71) is used to determine:

A. The period of availability of a specific appropriation type

✓ **B. A target profit/fee rate on negotiated defense contracts based on risk, contract type, and capital employed**

C. The contractor's overhead rate pool structure

D. The maximum labor categories allowed in a proposal

WHY

The Weighted Guidelines Method scores five factors (performance risk, contract type risk, facilities capital employed, cost efficiency, and independent R&D) to produce a target profit percentage. It ensures profit matches the risk and investment a contractor bears. Higher risk contracts earn higher fee rates. Government negotiators use this analysis to ensure profit is fair but not excessive.

Fringe benefits are applied to which cost base in a standard defense contractor cost buildup?

A. Total Contract Value

B. All indirect costs

✓ **C. Direct Labor Rate: fringe is a percentage of the employee's direct labor**

D. G&A base only

WHY

Fringe benefits (FICA, health insurance, retirement contributions, vacation/sick leave accrual, workers' comp) are applied as a percentage of direct labor dollars. They represent the employer's cost of employee benefits above base salary. A fringe rate of 32% on a \$70/hr DLR adds \$22.40/hr, representing the employer's share of the employee's total compensation package.

Build a wrap rate from scratch. Order these cost elements in the correct sequence of application:**CORRECT ORDER**

1. Direct Labor Rate (DLR): base hourly rate for the labor category|||Correct. The DLR is the foundation of cost-type contract pricing. It represents the base hourly compensation for a specific labor category (e.g., Senior Systems Engineer: \$85/hr). All indirect costs (fringe, overhead, G&A) and fee are applied on top of the DLR.
2. Fringe Benefits: applied to DLR (health, retirement, FICA, leave)|||Correct. Fringe is an indirect cost pool applied to direct labor. It typically includes FICA (7.65%), health/dental/vision insurance, 401k match, paid leave, and workers' compensation. Fringe rates for large contractors run 25-40% of direct labor.
3. Overhead: applied to direct labor within a business unit|||Correct. Overhead covers costs of running a business division not directly attributable to a specific contract: division management, facility costs, IT infrastructure, training. Overhead rates vary by business unit and are subject to DCAA Rate Agreements.
4. G&A: applied to Total Cost Input (all of the above)|||Correct. General and Administrative (G&A) expense covers enterprise-wide costs: CEO, CFO, legal, HR, corporate marketing. G&A is typically applied to Total Cost Input. G&A rates run 10-20% for large defense contractors and is the last indirect rate layer before fee.
5. Fee / Profit: applied to total estimated cost as final layer|||Correct. Fee (on cost-type contracts) or profit (on fixed-price) is negotiated between the CO and contractor. FAR 15.404-4 provides fee guidelines. For CPFF contracts, fee is fixed at negotiation. For CPAF, additional fee can be earned based on performance.

WHY

Wrap rate builds bottom-up: Direct Labor is the base → Fringe is applied to DLR → Overhead is applied to direct labor (or labor+fringe depending on contractor structure) → G&A is applied to total cost input (all direct + overhead costs) → Fee is the final layer applied to total estimated cost. This sequence matches FAR Part 31 cost structure.

Match each indirect rate type to its correct description:**CORRECT MATCHES**

Fringe Benefits	Health, retirement, FICA, % of direct labor
Overhead Rate	Division-level indirect costs: facilities, supervision, indirect labor
G&A Rate	Corporate-wide costs: executive mgmt, legal, finance, % of total cost input
FPRA	Pre-negotiated indirect rates used across all forward-priced proposals

WHY

Understanding the structure: Fringe is individual-employee-level benefits. Overhead is division-level indirect cost. G&A is enterprise-level corporate cost. CAS ensures consistent accounting across all of these. FPRAs lock these rates in advance for government proposals.

DEFENSE FINANCE & BUDGETING · LESSON 7

DCAA & DCMA Oversight

20 min

Understand the roles, authorities, and tools of DCAA and DCMA. And how to work with them effectively to protect your program and the taxpayer.

The Auditors and the Administrators

Most people in defense contracting hear DCAA or DCMA and feel a mild sense of dread. Wrong reaction. DCAA (Defense Contract Audit Agency) looks at the money: are the costs you're claiming allowable, allocable, and reasonable? DCMA (Defense Contract Management Agency) looks at the work: are you delivering what the contract says, and is your program data reliable? Think of DCAA as the financial auditor and DCMA as the performance monitor. If you're keeping accurate records, spending on legitimate costs, and delivering what you promised, these agencies are your allies, not your adversaries.

THE ONE-LINE VERSION

DCAA is the accountants who make sure the money math is right. DCMA is the contract cops who make sure contractors deliver what they promised. If your issue is a number, you are dealing with DCAA. If your issue is a delivery, a schedule, or a quality problem, you are dealing with DCMA.

Two Agencies, Two Missions: Both Critical to Your Program

Every major defense PM operates within a three-way oversight structure: the PCO (Procuring Contracting Officer) awarded the contract, DCMA administers it day-to-day, and DCAA audits the financial side. Many new PMs confuse DCAA and DCMA. They have different missions, different authorities, and different relationships with your program. Understanding what each can and cannot do determines whether you use them as effective oversight tools or accidentally create friction that slows your program down.

DCAA vs. DCMA: Side-by-Side Comparison

DCAA: What They Actually Audit

DCAA does not audit every contractor. It prioritizes its limited resources on the highest-risk contracts. The major audit types every PM should understand: (1) Pre-Award Surveys. Before award of a cost-reimbursable contract, DCAA may audit the contractor's accounting system to confirm it meets FAR requirements. An "inadequate" accounting system can block contract award. (2) Incurred Cost Audits. After each fiscal year, DCAA audits the contractor's ICS to verify actual indirect rates and ensure only allowable costs were billed. These audits often take years to complete, creating significant cost uncertainty. (3)

Forward Pricing Rate Audits. DCAA audits proposed indirect rates to support FPRA negotiations. (4) Truth in Negotiations Act (TINA) / Defective Pricing Audits. Post-award audits to verify that cost or pricing data submitted before award was accurate, current, and complete.

THE ACCOUNTING SYSTEM ADEQUACY FINDING: A PM'S RED FLAG

When DCAA finds a contractor's accounting system "inadequate," it is a program emergency. An inadequate finding means the system does not properly identify, accumulate, and segregate costs by contract. Which means you cannot trust the costs being billed to your program. In extreme cases, the ACO may suspend contractor billing or require withholds (typically 5-10% of billings) until the system is corrected. If your contractor has an open inadequate accounting system finding, flag it to your PCO immediately and track the remediation plan.

DCAA Audit Universe: Major Audit Types

DCMA: Your Government Representative at the Contractor's Facility

DCMA is the operational arm of government contract oversight. On major programs, DCMA assigns a resident Administrative Contracting Officer (ACO) and team directly to the contractor's facility. This team monitors schedule, quality, property management, and contract compliance on a daily basis. DCMA's ACO has the authority to: consent to or disapprove major subcontracts, accept or reject deliverables, issue cure notices, issue show cause letters before termination, and manage Government-Furnished Property (GFP). For PMs, the ACO is your proxy at the contractor. Your most direct source of real-time performance intelligence.

DCMA Key Functions and PM Interactions

DFARS Business Systems: The Six Systems DCMA and DCAA Watch

DFARS 252.242-7005 (Contractor Business Systems) identifies six contractor business systems that are subject to government review and withholding if found "significant deficiencies": (1) Accounting System (DCAA audits), (2) Estimating System (DCAA audits), (3) Earned Value Management System. EVMS (DCMA validates), (4) Purchasing System (DCMA reviews), (5) Material Management and Accounting System. MMAS (DCMA), (6) Property Management System (DCMA). When DCAA or DCMA finds a "significant deficiency" in any of these systems, the ACO can withhold 5% of contract billings (up to 10% total). As a PM, a withheld billing is cash flow risk for your contractor and a potential harbinger of deeper systemic problems.

DO NOT OBSTRUCT DCAA ACCESS

DCAA auditors have the right to access contractor records, employees, and facilities under FAR 52.215-2 (Audit and Records. Negotiation). A contractor that limits, delays, or obstructs DCAA access is committing a serious FAR violation. Equally important: as a government PM, never pressure a contractor to limit DCAA access to manage a schedule or cost outcome. This constitutes improper interference with an audit. A federal law violation with serious career and legal consequences. If a contractor claims they cannot provide records to DCAA, contact your PCO and legal office immediately.

WORKING WITH YOUR ACO: PRACTICAL TIPS

The most effective PMs build a strong relationship with their DCMA ACO. Best practices: (1) Include the ACO in your monthly program reviews. They are your government ally at the contractor. (2) Read every DCMA report your ACO produces. They often contain early warnings before IPMR data shows problems. (3) Ensure your contract includes appropriate DCMA access clauses (FAR 52.246-2, 52.242-4). (4) Do not go around your ACO to address contractor performance issues. Work through them. (5) For CPARS, get the ACO's performance input before finalizing your assessment. Their daily visibility is invaluable.

INTERMEDIATE**Working the DCAA/DCMA Relationship to Your Advantage**

Mid-career PMs who treat DCAA and DCMA as adversaries are leaving oversight resources on the table. Both agencies work for the government. Their findings protect your program, not the contractor. Establish a quarterly touchpoint with your ACO and DCAA auditor: share your program priorities, the highest-risk WBS elements, and any contractor behaviors that concern you. DCAA can target its audit resources to your hot spots if you tell them where to look. Request a floor check audit when your contractor reports a staffing surge that isn't reflected in performance. Ask DCMA to review the contractor's subcontract management practices when you see a subcontractor performance problem the prime isn't addressing. These agencies have authorities you don't. Use them.

Contractor PMs: DCAA showing up is not an emergency. It's normal. If you've got a compliant accounting system, accurate timecards, and you're not billing unallowable costs, a DCAA audit is just a process. The PMs who panic are usually the ones who haven't been keeping clean books. The one thing that actually causes problems is being slow or uncooperative with access. DCAA has the right to your records under FAR 52.215-2. Stonewall them and it escalates fast, and it ends badly. Open the door, get your finance team in the room, and answer the questions honestly. Clean records, transparent answers, no drama.

ADVANCED**WITHHOLDING PAYMENTS AND DISALLOWING COSTS: WHEN AND HOW**

Senior PMs have mechanisms to financially pressure non-performing contractors that junior PMs rarely use. For cost-type contracts, DCAA can recommend disapproval of incurred cost submissions. Triggering a withhold of up to 5% of billed costs pending resolution. For fixed-price contracts with performance milestones, you can withhold milestone payments when deliverables fail to meet contract requirements. The key: document the basis for the withhold in writing, citing specific contract clauses (FAR 52.232-7 for cost; the specific milestone clause for FP). Contractors will push back aggressively and threaten REAs. That's acceptable. What's not acceptable: using withholds as negotiating leverage for contract modifications (that's coercion). Withholds must be based on legitimate contract performance deficiencies, not as a tool to extract concessions.

Key Terms

DCAA

Defense Contract Audit Agency. The DoD agency responsible for auditing contractor accounting systems, cost proposals, and incurred cost submissions. Provides independent financial advice to contracting officers.

DCMA

Defense Contract Management Agency. The DoD agency responsible for contract administration, quality assurance, and contractor performance oversight on major contracts. DCMA is the government's "eyes and ears" at contractor facilities.

ACO

Administrative Contracting Officer. A DCMA contracting officer delegated authority to administer specific aspects of a contract (billing, consent to subcontract, property management).

QAR

Quality Assurance Representative. A DCMA official who monitors contractor quality processes and product acceptance at the contractor's facility.

EVMS Compliance

DCMA validates that contractor EVMS meets ANSI/EIA-748 criteria before government accepts it for IPMR reporting. A non-compliant EVMS is a major program risk.

Accounting System Adequacy

DCAA determination that a contractor's accounting system properly identifies, accumulates, and reports costs in accordance with FAR Part 31 and CAS. Required before award of cost-reimbursable contracts.

Billing System

The contractor's system for submitting invoices to the government. DCAA audits billing systems for accuracy and compliance.

Estimating System

The contractor's system for preparing cost proposals. DCAA audits estimating systems to ensure proposed costs are reasonable and consistent with actuals.

CIPR

Contractor Insurance and Pension Review. DCAA audit verifying that insurance and pension costs charged to government contracts are allowable and allocable.

Provisional Billing Rates

Interim indirect rates approved by the ACO/DCAA for billing purposes while actual rates are being finalized in the ICS process.

Knowledge Check

What is the primary mission of DCAA?

- A. Administer contracts and oversee contractor performance at contractor facilities
- ✓ **B. Provide independent financial auditing of contractor accounting systems and cost proposals**
- C. Award defense contracts and negotiate contract modifications
- D. Certify contractor EVMS compliance with ANSI/EIA-748

WHY

DCAA's mission is financial auditing. It audits contractor accounting systems, cost proposals, and incurred cost submissions to protect the government's financial interests. DCAA does not administer contracts (that is DCMA's ACO) and does not have contract award authority (that is the PCO). EVMS certification is DCMA's function.

An Administrative Contracting Officer (ACO) from DCMA has the authority to:

- A. Award new contracts independently
- ✓ **B. Withhold contractor billings for EVMS deficiencies, consent to subcontracts, and accept deliverables**
- C. Conduct financial audits of contractor cost proposals
- D. Authorize Nunn-McCurdy breach notifications

WHY

The ACO has direct contract administration authority. Including consent to subcontract decisions, GFP management, billing approval, cure notices, and deliverable acceptance. The ACO can withhold up to 5% of billings (10% maximum total) when contractor business systems have significant deficiencies. DCAA conducts audits (separate from ACO), and PCOs award contracts.

A DCAA "inadequate" accounting system finding on your prime contractor most immediately means:

A. The contract must be terminated for default

✓ B. Cost billings may not be trustworthy; the ACO may withhold a portion of billings until the system is corrected

C. The contractor must rebid the contract

D. DCMA takes over all contractor financial management

WHY

An inadequate accounting system finding means DCAA cannot rely on the contractor's cost data. Which means billed costs may not accurately reflect what was charged to your contract. The ACO typically implements a billing withhold (5% to 10%) pending remediation. The PM should elevate this to the PCO and monitor the corrective action plan closely.

Which of the six DFARS Business Systems does DCMA validate (not DCAA)?

A. Accounting System

B. Estimating System

✓ C. Earned Value Management System (EVMS)

D. Billing System

WHY

DCMA is responsible for EVMS validation. Confirming the contractor's EVM system meets ANSI/EIA-748 criteria. DCAA audits the Accounting System, Estimating System, and Billing System. The split reflects each agency's expertise: DCMA understands program performance systems; DCAA understands financial accounting systems.

Under DFARS 252.242-7005, when DCAA or DCMA finds a "significant deficiency" in a contractor business system, the maximum billing withhold is:

A. 2%

B. 5%

✓ C. 10%

D. 25%

WHY

DFARS 252.242-7005 allows withholds of 5% per deficient system, up to a maximum of 10% total across all systems. These withholds are intended to incentivize rapid system correction while preserving contractor cash flow to the extent possible. Withholds are lifted when the ACO approves a corrective action plan or confirms the deficiency has been resolved.

An Incurred Cost Submission (ICS) is filed by contractors to:

A. Propose rates for the upcoming fiscal year

✓ B. Report actual indirect cost rates for a completed fiscal year, enabling reconciliation with billed rates

C. Submit progress payments for completed milestones

D. Certify EVMS compliance to DCMA

WHY

The ICS (filed within 6 months after fiscal year end) reports actual indirect rates for the year. DCAA audits these against the provisional rates billed to government contracts. If the contractor over-billed (provisional rates higher than actuals), the government gets a credit. If under-billed, the government owes additional cost on cost-reimbursable contracts. ICS audit backlogs of 5-10 years create significant financial uncertainty.

A DCMA Quality Assurance Representative (QAR) at a contractor facility primarily performs:

A. Financial audits of contractor billing records

✓ **B. Oversight of contractor quality processes, product inspections, and delivery acceptance at contractor facilities**

C. Negotiation of contract modifications

D. Performance of independent cost estimates

WHY

QARs are DCMA's quality oversight representatives resident at contractor facilities. They witness testing, inspect deliverables, monitor quality management systems, and sign off on acceptance. Their daily presence gives the government real-time visibility into product quality. Long before a PM receives formal reports or delivery.

The Truth in Negotiations Act (TINA) / Defective Pricing audit by DCAA is triggered when:

A. Any contract over \$350K is awarded

✓ **B. A contract over \$2M required the submission of certified cost or pricing data before award**

C. A contractor files an ICS late

D. A Nunn-McCurdy breach is declared

WHY

TINA (codified as 41 U.S.C. § 3502) requires contractors to submit certified cost or pricing data for contracts exceeding the TINA threshold, raised to \$10M for contracts awarded after June 30, 2026 under the FY2026 NDAA (where competition and other exceptions do not apply). Post-award, DCAA may audit whether that data was accurate, current, and complete at the time of submission. If data was defective (incorrect or outdated), the government is entitled to a price reduction equal to the amount overpaid.

Which statement about DCAA access to contractor records is correct?

A. Contractors may limit DCAA access to protect proprietary information

✓ **B. DCAA has the right under FAR 52.215-2 to access all relevant records; obstructing this access is a FAR violation**

C. DCAA access requires a court order for private contractor records

D. DCAA can only access records voluntarily provided by the contractor

WHY

FAR clause 52.215-2 (Audit and Records, Negotiation). Included in negotiated contracts. Gives the government (and DCAA) the right to examine and audit all records related to the contract. Contractor obstruction of DCAA access is a serious FAR violation. PMs must ensure contractors understand their audit access obligations and escalate access disputes immediately to the PCO and legal.

In the context of DFARS Business Systems, a "Purchasing System" review by DCMA focuses on:

A. The contractor's system for purchasing commercial supplies

✓ **B. The contractor's system for acquiring subcontracted work and materials: ensuring competition, cost analysis, and flowdown of contract requirements**

C. The contractor's enterprise resource planning (ERP) system

D. The contractor's EVMS purchasing module

WHY

DCMA's Purchasing System review examines how the contractor manages its supply chain. Whether it properly competed subcontracts, performed cost/price analysis on non-competitive subs, and flowed down required government contract clauses to subcontractors. Poor purchasing system controls can result in overbilled subcontract costs, undisclosed subcontractor teaming, and failure to meet small business subcontracting goals.

Match each oversight function to the correct agency:**CORRECT MATCHES**

Accounting system adequacy audit	DCAA
EVMS validation and compliance	DCMA
Incurred Cost Submission audit	DCAA
Deliverable acceptance and quality oversight	DCMA

WHY

DCAA is exclusively financial auditing. Accounting systems, cost proposals, and incurred costs. DCMA is contract administration and performance oversight. ACO authority, EVMS validation, quality oversight, and CPARS. Neither agency awards contracts (that is the PCO) or certifies programs (that is the SAE/MDA).

Order these DCAA audit activities from the earliest in the contract lifecycle to the latest:**CORRECT ORDER**

1. Pre-Award Accounting System Survey (before cost-type contract award)|||Correct. Before a contractor can receive a cost-type contract, DCAA reviews whether their accounting system can adequately track and segregate costs by contract. An inadequate accounting system is a disqualifying condition. DFARS 252.242-7006 defines the 18 criteria for an adequate accounting system.
2. Forward Pricing Rate Audit (supporting FPRA before/at award)|||Correct. DCAA audits proposed indirect rates (fringe, overhead, G&A) and direct labor rates to support Forward Pricing Rate Agreements (FPRAs). These rates are then used as the basis for pricing all subsequent proposals under that contractor.
3. Provisional Billing Rate Review (during contract performance)|||Correct. During performance, contractors bill at Provisional Billing Rates (PBRs), estimated indirect rates, until actual rates are determined at year end. DCAA reviews these rates periodically to ensure they are reasonable and consistent with actual cost trends.
4. Incurred Cost Submission (ICS) Audit (after each fiscal year end)|||Correct. After each fiscal year, contractors submit an Incurred Cost Submission (ICS), an accounting of all actual costs incurred on government contracts. DCAA audits the ICS to determine final actual indirect rates, which may result in adjustments to prior billings.

WHY

Pre-award surveys happen before contract award. Forward pricing rate audits support proposal pricing before or at award. Provisional billing rate reviews happen during contract performance. Incurred cost submission audits happen after each fiscal year ends. TINA/defective pricing audits can happen anytime post-award, but are often triggered years after award.

DEFENSE FINANCE & BUDGETING · LESSON 8

CPAF, Burn Rate & Award Fee

55 min

How CPAF contracts actually pay you: the four award-fee factors, how burn rate protects your cost score day to day, and how every period rolls into a CPARS record that follows your company into the next recompete.

Your technical decisions are financial decisions.

On a CPAF contract, every staffing call, every deliverable, every cost approval shows up on the company income statement. You are not just managing a project. You are running a business unit where your performance score is the profit lever.

THE NUMBER THAT MATTERS

Booz Allen Hamilton earns ~98% of its revenue from government contracts. Lockheed Martin ~99%. When your contract wins a high award fee, that flows directly to operating income, earnings per share, and analyst confidence. Your program review is a Wall Street event.

How a CPAF Contract Pays You

The Four Factors That Score Your Profit

So How Do You Actually Protect That Cost Control Score?

You now know Cost Control is one of the four factors the government scores you on, and that the exact weighting depends on your specific contract's Award Fee Determination Plan. Knowing the score exists is one thing. Protecting it day to day, before the evaluation period even closes, is another. That is what burn rate tracking is for.

Burn Rate: The Speed Limit Nobody Posts

Award Fee Ends Each Period. CPARS Outlives the Contract.

Everything so far, the four factors, the rating table, burn rate, resets every evaluation period. There is a second scoreboard running underneath all of it that never resets: CPARS. Every period you just learned to protect becomes a permanent performance record that follows your company into the next recompete, and into every other contract your company bids on afterward.

CPARS SCORES DETERMINE FUTURE WORK

Your Contractor Performance Assessment Reporting System ratings are the number one factor government agencies use when deciding who wins re-compete awards. Strong CPARS = backlog growth. Weak CPARS = your company loses the follow-on. This is not abstract.

Now, the Ways PMs Actually Lose This Money

You know how the score works, how to protect the cost number day to day, and what is riding on it beyond this one contract. These are not hypotheticals. They are the most common ways PMs leave money on the table anyway, and each one maps directly back to a factor or a scoreboard you just learned.

Six Ways PMs Leave Money on the Table

Key Terms

CPAF

Cost-Plus Award Fee. A contract type where the government reimburses 100% of allowable costs and pays two layers of fee: a small fixed base fee (max 3%) and a variable award fee earned through performance ratings.

Award Fee

The variable, performance-linked portion of contract profit. Earned only for above-satisfactory performance across rated factors (Technical, Cost, Schedule, Management). Unsatisfactory performance earns \$0 regardless of other factors.

Base Fee

The fixed, guaranteed floor of profit on a CPAF contract. Capped at 3% of estimated contract cost under DFARS 216.405-2. Paid regardless of performance scores.

Wrap Rate

The cost multiplier applied to every dollar of direct labor: $\text{Billed Cost} = \text{Base Salary} \times (1 + \text{Fringe}) \times (1 + \text{Overhead}) \times (1 + \text{G\&A})$. A single employee at \$100K base salary can bill \$200K+ to the government after wrap rates are applied.

Burn Rate

The actual monthly cost spend rate on a contract. Finance calculates it weekly. If burn rate exceeds the remaining funded value, all work must stop. Spending beyond obligated funding violates the Anti-Deficiency Act (ADA), a federal offense.

Labor Utilization

$\text{Direct Hours} \div \text{Total Available Hours}$. The percentage of your team's time charged to direct (billable) work. Target is 75–85%. Below 70% inflates overhead rates company-wide; above 90% risks burnout and quality degradation.

Anti-Deficiency Act (ADA)

Federal law prohibiting government contractors from spending beyond officially obligated funding. An ADA violation is a federal criminal offense. The funded ceiling is set by contract modification and ACRN. Never exceed it.

EAC (Estimate at Completion)

Finance's projection of total contract cost by end of performance, based on current burn rate and remaining scope. Used to detect funding gaps before they become crises. PMs should sync with Finance on EAC weekly.

ODCs (Other Direct Costs)

Non-labor direct costs charged to a contract: travel, equipment, materials, training. High-risk cost area. Require pre-approval and documented contract nexus. Non-mission travel and unrelated training are waste that drains award fee.

DCAA

Defense Contract Audit Agency. Audits contractor indirect rates (fringe, overhead, G&A) and cost pools to ensure compliance with FAR Part 31. Unallowable costs (entertainment, alcohol, image advertising) must be excluded from all pools.

DEFENSE FINANCE & BUDGETING · LESSON 9

Revenue Recognition

14 min

Learn what revenue recognition means, why how you describe your work determines how much your company is worth, and how shifting from logistics to engineering changes your financial picture.

What Revenue Recognition Actually Means

Revenue recognition is the accounting rule that decides how much of a contract's value your company actually gets to count as revenue on its books. It sounds like an accounting detail, but it is the difference between your company looking like a \$50M business or a \$5M business off the exact same contract, depending on whether you are classified as a Principal (you book the full contract value) or an Agent (you book only your fee). Most PMs never think about how revenue gets reported. The way your company describes your work on paper determines how much the business is worth. Two companies can do identical work on the same DoD contract and end up with wildly different financial profiles depending on whether they are classified as a Principal or an Agent.

Principal vs. Agent: The \$9.5M Difference

The 5-Step Revenue Test

To recognize revenue as a Principal under ASC 606, your company must satisfy five steps. As a PM, your job is to make sure the work you manage is documented in a way that passes each step.

The 5-Step Revenue Recognition Framework

FROM LUCAS

I've sat in rooms where a company was pitching to a potential acquirer and the first question was "are you booking gross or net?" If the answer is net, the valuation conversation changes immediately. Your CPARS might be perfect, your team might be excellent, but on paper you're a pass-through. That matters when you're trying to grow, borrow money, win larger contracts, or sell the company.

The Hardware Dump Problem

One of the most common traps in defense contracting is the hardware dump. A Statement of Work that only asks for hardware delivery with no engineering work described. If your SOW just says "deliver 50 servers," you're an Agent. But if that same SOW says "deliver, configure, harden, integrate, and test 50 servers per the government's cybersecurity requirements," you're a Principal.

From Logistics to Engineering: The Language Shift

IF IT IS NOT DOCUMENTED, IT DID NOT HAPPEN

From an auditor's perspective, verbal work and undocumented engineering are invisible. If your team spent 300 hours integrating a system but the SOW does not describe integration as a performance obligation, that work does not support a Principal classification.

One sentence to teach your team: **the government pays the same amount either way, but a Principal reports the whole contract as revenue and an Agent reports only its fee, and that difference decides what the company is worth.**

INTERMEDIATE

The Control Test: Five Indicators Auditors Use

INDICATOR	POINTS TO PRINCIPAL	POINTS TO AGENT
Primary responsibility for fulfilling the promise	Your company is accountable to the government for the whole deliverable	The subcontractor or supplier is really the one on the hook
Inventory or performance risk	You bear the risk if the work fails or the goods are rejected	Risk sits with the supplier; you just pass the item through
Pricing discretion	You set the price to the government	Price is effectively set by the supplier and marked up
Integration and modification	Your people configure, integrate, test, or change the item before delivery	The item arrives and leaves untouched
Direction of the work	You issue technical direction and accept or reject subcontractor output	You forward requests and invoices

What Changes When You Move From Agent to Principal

On the same \$10M contract, an Agent booking a 5% fee reports \$500K of revenue at a 100% gross margin. A Principal reports \$10M of revenue at, say, a 12% gross margin. Profit in dollars can be similar. Everything else is different. Revenue is twenty times larger, which changes the company's size category with banks, buyers, and customers. Gross margin percentage drops sharply, which finance must explain to anyone comparing the two years. Receivables and unbilled balances grow, because the whole contract value now flows through the books, which changes working capital needs. Loan covenants based on revenue or leverage ratios move. And the company now needs the processes to actually control the work it is now

reporting: subcontract management, quality acceptance, technical direction. Moving from Agent to Principal is not a paperwork change. It is a change in what the company does.

Writing the Proposal and SOW Response to Support Principal Treatment

The classification is decided by facts, but the facts have to be visible in the contract documents. When responding to a SOW or writing a PWS, make the work your company controls explicit.

- **Name the integration and configuration tasks** "Configure, harden, integrate, and test" is a performance obligation. "Deliver" is a hand-off.
- **State that your company accepts and warrants the whole solution** A warranty or acceptance-test obligation on the integrated system is strong evidence of primary responsibility.
- **Describe technical direction of subcontractors** If your engineers direct the sub's work and accept its output, say so and then do it.
- **Show your pricing discretion** Pricing built from your own labor, subcontract costs, and rates, not a supplier quote plus a percentage.
- **Keep the evidence after award** Technical direction records, acceptance reports, and integration test results are what the auditor reads.

ADVANCED

Mixed Contracts: Principal on Some CLINs, Agent on Others

The test is applied per performance obligation, not per contract. A single contract can have a services CLIN where the company is clearly a Principal (its own engineers, its own risk) and a hardware CLIN where it is an Agent (pass-through of a manufacturer's product with no modification). The company books the services CLIN gross and the hardware CLIN net. Finance needs the PM to tell them which CLINs involve real control and which do not, because the contract documents alone often do not make it obvious. A PM who says "it is all one contract, book it all gross" is creating an audit finding. A PM who can walk through each CLIN and say what the company actually does on it is giving the auditor exactly what the standard asks for.

The Revenue Memo: What Finance Needs From the Program

For any significant contract, the company's accountants write a revenue recognition memo that documents the Principal-versus-Agent conclusion and the over-time-versus-point-in-time conclusion. External auditors read it. The memo needs facts only the program can supply: what the company's staff actually do on each performance obligation, who accepts subcontractor work, who bears the risk of rework, how pricing was built, and what happens if the deliverable fails. Senior PMs should expect to be interviewed for this memo at contract start and again whenever scope changes materially. Answering it accurately is part of the job, and answering it loosely is how a company ends up restating revenue.

THE CONVERSATION WITH AN ACQUIRER

The first question is "gross or net?" The second is "show me why." A buyer's diligence team will pick the largest contracts, read the SOWs, look at the org chart, and ask who really does the work. If the answer is a subcontractor and a project manager who forwards invoices, the revenue gets restated to net in the buyer's model, and the price follows. If the answer is documented integration, direction, and acceptance by the company's own people, the revenue stands. The Business of Defense Contracting module goes deeper on both the accounting and the valuation.

BUILD ON THIS

- **Revenue Recognition II** Over-time recognition and how the EAC drives revenue
- **Pass-Through vs. Value-Add** The government's own version of the Principal/Agent question
- **How GovCon Companies Are Valued** What gross vs. net does to the sale price

Key Terms

ASC 606

The FASB accounting standard governing revenue recognition. Determines when and how revenue is recorded, and whether the full contract value or just the fee is recognized as revenue.

Principal

A party that controls delivery of goods or services and bears performance risk. Books gross revenue under ASC 606.

Agent

A party that arranges delivery but does not control it or bear risk. Books only the fee as revenue.

Performance Obligation

A documented promise to deliver a distinct good or service. Must be specified in the contract to support revenue recognition.

Value-Add

Technical work, such as integration, hardening, configuration, or testing, that transforms a product and supports Principal classification.

GAAP

Generally Accepted Accounting Principles. The accounting rules your company's financials follow. ASC 606 is part of GAAP.

Gross vs. Net Revenue

Gross = full contract value. Net = fee only. The difference determines how large and valuable your company appears to investors and lenders.

Knowledge Check

Under ASC 606, a contractor who takes ownership of delivery, adds technical value, and bears performance risk is classified as:

- A. An Agent
- B. A Subcontractor

✓ C. A Principal

- D. A Pass-Through Entity

WHY

A Principal controls the delivery of a good or service and bears the risk of delivery. They book gross revenue. The full contract value. An Agent arranges delivery but does not control it, and books only the net fee.

A SOW says "Deliver 100 laptops to the government." With no other engineering work described, this contractor would most likely be classified as:

- A. A Principal: hardware delivery is always gross revenue

✓ B. An Agent: no value-add is documented

- C. A Subcontractor under ASC 605
- D. A Principal if the laptops cost more than \$150K

WHY

Without documented engineering work, a hardware delivery SOW describes a pass-through arrangement. The contractor books only their fee as revenue. Not the gross hardware cost.

A PM ensures the SOW includes "security hardening, system integration, and lifecycle testing." The primary financial benefit is:

- A. It increases the contract ceiling
- ✓ **B. It supports Principal classification and gross revenue recognition**
- C. It qualifies the contract for CPIF fee structure
- D. It satisfies Nunn-McCurdy threshold requirements

WHY

Documenting engineering work as explicit performance obligations supports booking gross contract revenue rather than just the fee. Directly improving the company's reported financials.

Which of the following most strongly indicates Principal treatment under ASC 606?

- A. The company signed the prime contract
- ✓ **B. The company's engineers integrate and test the subcontracted components and accept or reject the sub's work**
- C. The company forwards the supplier's invoice with a markup
- D. The contract value exceeds \$10M

WHY

Control before transfer is shown by integration, direction, and acceptance of the work, not by holding the contract.

A company moves from Agent to Principal on a \$10M contract with a 5% fee. What happens to reported gross margin percentage?

- A. It rises
- B. It stays the same
- ✓ **C. It falls sharply, because the full contract value is now revenue**
- D. It becomes zero

WHY

As an Agent, \$500K of fee is 100% margin. As a Principal, the same profit sits on \$10M of revenue, so the percentage drops even though dollars are similar.

A contract has a services CLIN performed by the company's own staff and a hardware CLIN that passes a manufacturer's product through unmodified. How is revenue classified?

- A. All gross, because it is one contract
- B. All net, because part of it is pass-through

✓ C. Per performance obligation: services gross, hardware net

- D. Whichever the PM prefers

WHY

The Principal/Agent test is applied to each performance obligation. Mixed treatment on one contract is normal.

What is the purpose of a revenue recognition memo?

- A. To bill the customer

✓ B. To document the Principal/Agent and over-time conclusions with facts from the program for the auditors

- C. To set indirect rates
- D. To request a contract modification

WHY

The memo records the accounting conclusions and the facts behind them. The PM supplies the facts about who does what.

Which SOW language most supports Principal treatment?

- A. "Deliver 50 servers to Building 4"

✓ B. "Deliver, configure, harden, integrate, and acceptance-test 50 servers per the government's security baseline"

- C. "Provide servers as specified by the manufacturer"
- D. "Coordinate delivery of 50 servers"

WHY

Configuration, integration, and acceptance testing are performance obligations the company controls. Delivery alone is a hand-off.

Why would an acquirer restate a target's revenue from gross to net?

A. Because the target is too large

✓ B. Because diligence shows a subcontractor does the work and the company merely holds the contract

C. Because the contract is cost-type

D. Because CPARS ratings are low

WHY

Buyers test the Principal claim against who actually performs and controls the work. If control is absent, they model the revenue as net and price accordingly.

Match the indicator to the classification it supports:**CORRECT MATCHES**

Company bears risk if the deliverable fails	Principal
Company sets the price to the government	Principal
Company forwards the supplier invoice plus a percentage	Agent
Company issues technical direction and accepts sub output	Principal

WHY

Bearing performance risk, setting price, and directing the work point to Principal. Forwarding invoices with a markup points to Agent.

DEFENSE FINANCE & BUDGETING · LESSON 10

What Your CFO Sees

40 min

Plain English breakdown of gross margin, operating margin, contract profit, fee types, and DSO. The financial language that gets you into the room with leadership.

You know the mission. Do you know the money?

Most contractor PMs can tell you exactly how many deliverables are due this month, where the schedule stands, and what the government customer thinks of the program. But ask them what the gross margin is and you get a blank stare. That gap costs people promotions. Business unit leaders, executives, and proposal teams all speak the language of margin and fee. If you can cross that line, you stop being the person who executes the work and start being the person leadership brings into the room when decisions are made.

TWO SCORECARDS. ONE PROGRAM.

The government tracks your program on deliverables, schedule, and performance. Your company tracks it on margin, fee, and cash flow. Both are running at the same time. A program can be green on the government side and red on the business side. You need to know both.

Where Does \$100 Go on a Defense Contract?

How Each Fee Type Works

I was in a quarterly business review early in my career and the CFO asked about our DSO and gross margin variance against plan. I had no idea what he was talking about. The program was performing. Deliverables on time, government happy, no issues. But I could not speak to the financial health of the business unit. That was the moment I realized there was a whole other game being played above my head. The PMs who move up are the ones who can say: our gross margin is running two points ahead of plan because we right-sized the subcontractor team in Phase 2, and if we earn 85% of our award fee this quarter, we will finish the year at 9% operating margin. That is the language of leadership.

DSO: The Cash Gap Nobody Explains

THE MISTAKE MOST PMS MAKE

Treating technical performance and financial performance as the same thing. A program can be green on schedule and still be a problem for your business unit if costs are running over, DSO is high, or award fee is being lost.

Award Fee Is Not Just a Rating. It Is a Dollar Amount.

On a \$50M CPAF contract with a 10% award fee pool, the difference between earning 85% of the pool and earning 60% is \$1.25M in profit for the year. Scale that across 5 to 10 programs in a business unit and the difference between average and excellent award fee performance is the difference between making the annual plan and missing it.

THE CONVERSATION THAT CHANGES YOUR CAREER

Walk me through how your program financial performance connects to our business unit results this quarter. If you can answer that without hesitating, you are no longer just executing the work. You are a business partner. That is the promotion track.

INTERMEDIATE

Reading a Program P&L: One Contract, Top to Bottom

LINE	EXAMPLE (\$ FOR THE YEAR)	WHAT IT MEANS
Revenue	\$10,000,000	What the company recognized for this program (see Revenue Recognition)
Direct labor	\$4,200,000	Salaries of people charged to the contract
Fringe on direct labor	\$1,260,000	Benefits and payroll taxes at the fringe rate
Subcontracts and materials	\$1,500,000	What you paid others
Other direct costs	\$300,000	Travel, equipment, and similar
Gross margin	\$2,740,000 (27.4%)	Revenue minus all direct costs. The number leadership compares across programs
Overhead allocation	\$1,092,000	Overhead rate applied to direct labor plus fringe
G&A allocation	\$880,000	G&A rate applied to the cost base
Operating margin	\$768,000 (7.7%)	What is left for the company after every allocation. This is the number the CFO reports upward

Fee and Margin Are Not the Same Number

On a cost-type contract, fee is contractual: the fixed fee, the award fee earned, the incentive fee calculated. Margin is what the company actually keeps after everything the government does not pay for: unallowable costs, indirect rate variances, over-ceiling work, and unbilled time. A program can earn its full 8% fee and deliver a 5% operating margin because the business unit's overhead rate ran above the provisional rate all year. When leadership asks about margin and you answer with fee, you have answered a different question. Know both numbers and be able to explain the gap between them.

The Four Questions in Every Business Review

Quarterly business reviews at contractors follow a pattern. Walk in with the answers.

- **Are we on plan for revenue?** Recognized revenue against the annual plan for this program, with the reasons for any variance: staffing, funding timing, scope.
- **Are we on plan for margin?** Gross and operating margin against plan, and whether the gap is program-driven (labor mix, subs, unbilled work) or company-driven (indirect rates).

- **What is the cash position?** Unbilled balance, DSO, anything stuck in WAWF, any withholds. Leadership cares about cash as much as margin.
- **What changes the picture next quarter?** Award fee determination, option exercise, funding mods, EAC risks, key staff departures. The forward look is the part that gets you invited back.

ADVANCED

Why Your 10% Program Shows Up as 6% in the Roll-Up

Program margin and business unit margin are different views. Your program P&L shows margin after the allocations it was budgeted to carry. The business unit P&L adds everything that was not allocated to any program: bench labor above plan, B&P spend above plan, indirect rate variances, unallowable costs, and reserves finance sets aside for program risk. A unit with five programs each at 10% can report 6% because the unallocated layer ate four points. Senior PMs understand this, which is why they care about company-wide utilization and rate variance even when their own program is on plan. It is also why "my program is fine" is never a complete answer in a business review when the unit is missing plan.

Translating Program Status Into Financial Language

WHAT YOU WOULD NORMALLY SAY	WHAT LEADERSHIP HEARS	SAY THIS INSTEAD
"We are two weeks behind on the integration milestone"	Is revenue moving out of the quarter?	"The slip moves about \$300K of revenue from Q3 to Q4; EAC is unchanged"
"We lost our lead engineer"	Is there an EAC impact and a fee impact?	"Replacement onboarded; two weeks of overhead absorbed; no change to EAC; possible award fee comment"
"The customer keeps asking for extra reports"	Unbilled scope creep	"About 120 unfunded hours this quarter; I have raised an REA candidate with contracts"
"Award fee period went well"	How many dollars?	"We expect 88% of the \$400K pool, about \$50K above the booked estimate"
"Funding is late again"	Cash and unfunded backlog	"\$1.2M unfunded as of today; CO expects the mod in three weeks; we will hit LOF notification on the 20th if not"

WHAT SENIOR PMS DO DIFFERENTLY

They read their program P&L monthly and know gross margin, operating margin, and the gap between fee and margin. They translate every program event into revenue, margin, and cash before the business review asks. They understand the unallocated layer between their program and the business unit result. And they treat the forward look as the most important slide, because leadership already knows what happened last quarter.

BUILD ON THIS

- **Cash Is Not Revenue** The full cash story behind DSO
- **Indirect Rate Games** Why the roll-up margin is lower than your program margin
- **Why Q3 Was Bad** What an EAC change does to the quarter

Key Terms**Gross Margin**

Revenue minus direct costs. Does not include overhead or G&A. A 30% gross margin means for every \$1 billed, \$0.30 is left before paying company overhead.

Operating Margin

What is left after overhead and G&A are paid. Defense service companies typically target 8 to 12%.

Contract Profit

The fee earned on a specific contract after all allowable costs are paid. On cost-plus, profit equals fee. On fixed-price, profit equals price minus actual costs.

Fixed Fee

A set dollar amount paid to the contractor regardless of performance. Typically 7 to 10% on cost-plus contracts.

Award Fee

Variable fee earned through performance ratings from a government Award Fee Board. Your rating determines how much of the pool you keep.

DSO

Days Sales Outstanding. How many days between billing the government and receiving cash. Under 45 days is healthy. Above 60 means the company is fronting money it has not collected yet.

Unallowable Costs

Costs the government refuses to reimburse under FAR Part 31. Includes entertainment, lobbying, and advertising. DCAA will find them and require repayment.

EBITDA

Earnings Before Interest, Taxes, Depreciation, and Amortization. Used in M&A and corporate valuation. PE-backed contractors watch this closely.

Knowledge Check

A contract has \$10M in estimated costs and a \$1M fixed fee. Actual costs come in at \$9.5M. What does the contractor earn in fee?

- A. \$950K because fee adjusts to 10% of actual costs
- ✓ **B. \$1M because fixed fee does not change based on actual costs**
- C. \$500K because savings are split with the government
- D. \$1.5M because they keep the cost savings plus the fee

WHY

Fixed fee is set in dollars at award and does not change. The contractor earns \$1M regardless of whether costs were \$9.5M or \$10M. Cost savings go back to the government on CPFF contracts.

Your program has a 28% gross margin but only a 4% operating margin. What does this gap most likely mean?

- A. The program is highly profitable
- ✓ **B. Overhead and G&A are eating most of the gross profit**
- C. Revenue is being deferred to a future period
- D. The fee was negotiated too low

WHY

A 24-point gap between gross and operating margin means indirect costs are consuming almost all of the program contribution. This is common at large primes with heavy corporate overhead.

Your company DSO is 72 days. Why is this a problem even if the contract is performing fine?

- A. It means the government is disputing your invoices
- ✓ **B. The company is paying expenses today but collecting cash 72 days later**
- C. DCAA will flag this as a billing error
- D. It signals unallowable costs in the indirect pool

WHY

High DSO means cash is stuck in receivables. The company still pays salaries, benefits, and overhead every two weeks. Collecting 72 days after billing means you are effectively lending money to the government interest-free.

The Award Fee Board rates your program at 78% for the quarter. The fee pool for that period is \$500K. What does the contractor earn?

- A. \$500K because the full pool is always paid
- ✓ B. \$390K which is 78% of the \$500K pool
- C. \$78K because the percentage applies to total contract value
- D. \$0 because most contracts require 80% to qualify

WHY

78% of \$500K equals \$390K. Award fee scales directly with the rating. There is no universal pass/fail threshold.

Which of these costs is unallowable under FAR Part 31?

- A. Business travel to attend a program review
- B. Employee health insurance premiums
- ✓ C. A company-sponsored golf outing for government customers
- D. Subcontractor invoices for engineering support

WHY

FAR 31.205-14 explicitly makes entertainment costs unallowable. Golf outings, sporting events, and client dinners cannot be billed to a government contract even when government employees attend.

A program has \$10M revenue, \$7.26M of direct costs, and \$1.97M of overhead and G&A allocations. What are gross margin and operating margin?

- ✓ A. 27.4% and 7.7%
- B. 7.7% and 27.4%
- C. 35% and 10%
- D. 19.7% and 7.7%

WHY

Gross margin = $(\$10M - \$7.26M) \div \$10M = 27.4\%$. Operating margin = $(\$10M - \$7.26M - \$1.97M) \div \$10M = 7.7\%$.

A CPFF program earned its full 8% fee but reports a 5% operating margin. What most likely explains the gap?

A. The government underpaid

✓ **B. Unallowable costs, indirect rate variances, or unbilled work reduced what the company kept**

C. The fee was miscalculated

D. Revenue was recognized twice

WHY

Fee is contractual. Margin is what remains after costs the government does not reimburse. The gap between them is the company's problem, not the customer's.

Five programs in a business unit each report 10% margin, but the unit reports 6%. What is the most likely cause?

A. Accounting error

✓ **B. The unallocated layer: bench labor, B&P overruns, rate variances, and reserves not charged to any program**

C. The programs are overstating revenue

D. Award fee was lost

WHY

Costs not allocated to programs still hit the business unit P&L. Program margin and unit margin are different views of the same money.

Your integration milestone slipped two weeks into next quarter. What is the financial translation leadership wants?

A. "The customer is fine with it"

✓ **B. The revenue moving between quarters and whether the EAC changed**

C. The name of the engineer responsible

D. Nothing; schedule is not a financial topic

WHY

Leadership hears a schedule slip as a revenue timing question. Give the dollars and the EAC status.

Which item belongs in the "forward look" of a business review?

A. Last quarter's revenue

✓ B. The expected award fee determination, option exercise timing, and funding mod dates

C. The org chart

D. The contract number

WHY

Leadership already knows last quarter. The events that will change next quarter's revenue, margin, and cash are what they need from the PM.

Order these lines as they appear on a program P&L from top to bottom:

CORRECT ORDER

1. Revenue
2. Direct labor, fringe, subcontracts, ODCs
3. Gross margin
4. Overhead and G&A allocations
5. Operating margin

WHY

Revenue, then direct costs, then gross margin, then indirect allocations, then operating margin.

DEFENSE FINANCE & BUDGETING · MODULE ASSESSMENT

Module Assessment

Pass this to unlock the next skill level in the app.

Knowledge Check

Which appropriation type funds the development and testing of a new weapons system before production begins?

A. Operations & Maintenance (O&M)

B. Procurement

✓ C. Research, Development, Test & Evaluation (RDT&E)

D. Military Construction (MILCON)

WHY

RDT&E funds the research, development, test, and evaluation activities before a system enters production. It is a two-year appropriation and cannot be used for procurement of production-representative articles beyond a limited number of test articles.

Earned Value Management (EVM): if your program has a CPI of 0.85, what does this mean?

A. You are spending \$0.85 for every \$1.00 of planned work: ahead of budget

✓ B. You are getting \$0.85 of work for every \$1.00 spent: over budget

C. Your schedule is 15% behind plan

D. The contract has a 15% profit fee

WHY

CPI (Cost Performance Index) = EV / AC . A CPI of 0.85 means you are only getting 85 cents of earned value for every dollar spent. You are over budget. A CPI below 1.0 is a cost overrun indicator.

The PPBE cycle's "Programming" phase produces which key document?

A. Program Objective Memorandum (POM)

B. President's Budget (PB)

C. Future Years Defense Program (FYDP)

✓ D. Both A and C: the POM feeds into the FYDP

WHY

During Programming, each Military Department submits a POM to OSD. The approved programs are then reflected in the FYDP. The official 5-year financial plan. The POM IS the programming document; the FYDP is the output.

An Estimate at Completion (EAC) calculated as BAC / CPI represents which assumption?

A. All remaining work will be completed at the budgeted rate

✓ **B. Future work will continue at the same cost efficiency as work completed to date**

C. A specific re-estimate of remaining work was performed

D. The contract will be terminated for convenience

WHY

$EAC = BAC / CPI$ assumes future cost performance will mirror past cost performance. It is the most common EAC method used by DoD and tends to be the most accurate on programs with stable performance trends.

O&M (Operations & Maintenance) funding expires after how long?

✓ **A. 1 year**

B. 2 years

C. 3 years

D. 5 years

WHY

O&M is a one-year appropriation. It must be obligated within the fiscal year it is appropriated. Using it after expiration (expired funds) or for the wrong purpose violates appropriations law and can trigger an Anti-Deficiency Act (ADA) violation.

Which EVM metric directly measures schedule efficiency?

A. CPI (Cost Performance Index)

✓ **B. SPI (Schedule Performance Index)**

C. TCPI (To-Complete Performance Index)

D. VAC (Variance at Completion)

WHY

$SPI = EV / PV$. An SPI below 1.0 means you have accomplished less work than planned. You are behind schedule. Note: EVM-based SPI is a dollar-weighted measure, not a time-based one. SPI converges to 1.0 at program completion regardless of lateness.

A program manager discovers that O&M funds were used to buy a capital asset worth \$500K. This is most likely a violation of:

A. The Competition in Contracting Act

B. The Bona Fide Needs Rule

✓ C. The Purpose Statute (31 U.S.C. § 1301)

D. The Antideficiency Act

WHY

The Purpose Statute requires that appropriations be used only for the purposes for which they were appropriated. Buying capital assets with O&M funds violates this statute. Capital equipment should be funded with Procurement or RDT&E depending on the item.

TCPI (To-Complete Performance Index) is used to determine:

A. How efficiently the program has spent money to date

✓ B. The cost efficiency required on remaining work to meet a target (BAC or EAC)

C. Whether the current contract type is appropriate

D. The final profit margin the contractor will earn

WHY

$TCPI = (BAC - EV) / (BAC - AC)$ when calculated against BAC. A TCPI above 1.0 means remaining work must be performed more efficiently than past work. This is a warning sign. A TCPI above 1.1 is considered essentially unachievable without a re-baseline.